1) a)

$\int (2\cos 3x)^2 \mathrm{d}x = \int 4\cos^2(3x)\,\mathrm{d}x$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

$\int 4\cos^2(3x)\,\mathrm{d}x = \int (2 + 2\cos(6x))\mathrm{d}x$

$= 2x + \frac{1}{3}\sin 6x + c$

1) b)

$\int (2\sin 3x)^2 \mathrm{d}x = \int 4\sin^2(3x)\,\mathrm{d}x$

Since $\cos(2A) = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$

$\int 4\sin^2(3x)\,\mathrm{d}x = \int (2 - 2\cos(6x))\ \mathrm{d}x$

$= 2x - \frac{1}{3}\sin 6x + c$

1) c)

$\int (1 + \cos x)^2 \mathrm{d}x = \int (1 + 2\cos x + \cos^2 x)\mathrm{d}x$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

$\int (1 + 2\cos x + \cos^2 x)\mathrm{d}x = \int \left(\frac{3}{2} + 2\cos x + \frac{1}{2}\cos(2x)\right)\mathrm{d}x$

$= \frac{3}{2}x + 2\sin x + \frac{1}{4}\sin(2x) + c$

1) d)

$\int (1 - \sin x)^2 \mathrm{d}x = \int (1 - 2\sin x + \sin^2 x)\mathrm{d}x$

Since $\cos(2A) = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$

$\int (1 - 2\sin x + \sin^2 x)\mathrm{d}x = \int \left(\frac{3}{2} - 2\sin x - \frac{1}{2}\cos(2x)\right)\mathrm{d}x$

$= \frac{3}{2}x + 2\cos x - \frac{1}{4}\sin(2x) + c$

1) e)

Since $\cos(2A) = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$

$\int \left(\sin^2\left(\frac{x}{6}\right)\right)\mathrm{d}x = \int \left(\frac{1}{2} - \frac{1}{2}\cos\left(\frac{x}{3}\right)\right)\mathrm{d}x$

$= \frac{1}{2}x - \frac{3}{2}\sin\left(\frac{x}{3}\right) + c$

1) f)

$\int \left(2+\cos\left(\frac{x}{2}\right)\right)^2 \mathrm{d}x = \int \left(4+4\cos\left(\frac{x}{2}\right)+\cos^2\left(\frac{x}{2}\right)\right)\mathrm{d}x$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2}+\frac{1}{2}\cos(2A)$

$\int \left(4+4\cos\left(\frac{x}{2}\right)+\cos^2\left(\frac{x}{2}\right)\right)\mathrm{d}x = \int \left(\frac{9}{2}+4\cos\left(\frac{x}{2}\right)+\frac{1}{2}\cos x\right)\mathrm{d}x$

$= \frac{9}{2}x + 8\sin\left(\frac{x}{2}\right)+\frac{1}{2}\sin x + c$

2) a)

$\frac{1}{2}\int (4+3\cos\theta)^2 \mathrm{d}\theta = \frac{1}{2}\int (16+24\cos\theta+9\cos^2\theta)\mathrm{d}\theta$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2}+\frac{1}{2}\cos(2A)$

$\frac{1}{2}\int (16+24\cos\theta+9\cos^2\theta)\mathrm{d}\theta = \frac{1}{2}\int \left(16+24\cos\theta+9\left(\frac{1}{2}+\frac{1}{2}\cos(2\theta)\right)\right)\mathrm{d}\theta$

$= \frac{1}{2}\int \left(\frac{41}{2}+24\cos\theta+\frac{9}{2}\cos(2\theta)\right)\mathrm{d}\theta$

$= \frac{1}{2}\left(\frac{41}{2}\theta+24\sin\theta+\frac{9}{4}\sin(2\theta)\right)(+c)$

So that

$\frac{1}{2}\int_0^{2\pi}(4+3\cos\theta)^2\mathrm{d}\theta = \left[\frac{1}{2}\left(\frac{41}{2}\theta+24\sin\theta+\frac{9}{4}\sin(2\theta)\right)\right]_0^{2\pi}$

$= \left(\frac{1}{2}(41\pi+0+0)\right)-\left(\pi(0+0+0)\right)$

$= \frac{41}{2}\pi$ units^2

2) b)

$\frac{1}{2}\int (5+2\sin\theta)^2 \mathrm{d}\theta = \frac{1}{2}\int (25+20\sin\theta+4\sin^2\theta)\mathrm{d}\theta$

Since $\cos(2A) = 1-2\sin^2 A$ then $\sin^2 A = \frac{1}{2}-\frac{1}{2}\cos(2A)$

$\frac{1}{2}\int (25+20\sin\theta+4\sin^2\theta)\mathrm{d}\theta = \frac{1}{2}\int \left(25+20\sin\theta+4\left(\frac{1}{2}-\frac{1}{2}\cos(2\theta)\right)\right)\mathrm{d}\theta$

$= \frac{1}{2}\int (27+20\sin\theta-2\cos(2\theta))\mathrm{d}\theta$

$= \frac{1}{2}(27\theta-20\cos\theta-\sin(2\theta))(+c)$

So that

$\frac{1}{2}\int_0^{2\pi}(5+2\sin\theta)^2\mathrm{d}\theta = \left[\frac{1}{2}(27\theta-20\cos\theta-\sin(2\theta))\right]_0^{2\pi}$

$= \left(\frac{1}{2}(54\pi-20-0)\right)-\left(\frac{1}{2}(0-20-0)\right)$

$= 27\pi$ units^2

2) c)

$\frac{1}{2}\int \left(a(1-\cos\theta)\right)^2 \mathrm{d}\theta = \frac{1}{2}a^2\int (1-2\cos\theta+\cos^2\theta)\mathrm{d}\theta$

Since $\cos(2A) = 2\cos^2 A-1$ then $\cos^2 A = \frac{1}{2}+\frac{1}{2}\cos(2A)$

$\frac{1}{2}a^2\int (1-2\cos\theta+\cos^2\theta)\mathrm{d}\theta = \frac{1}{2}a^2\int \left(1-2\cos\theta+\frac{1}{2}+\frac{1}{2}\cos(2\theta)\right)\mathrm{d}\theta$

$= \frac{1}{2}a^2\int \left(\frac{3}{2}-2\cos\theta+\frac{1}{2}\cos(2\theta)\right)\mathrm{d}\theta$

$= \frac{1}{2}a^2\left(\frac{3}{2}\theta-2\sin\theta+\frac{1}{4}\sin(2\theta)\right)(+c)$

So that

$\frac{1}{2}\int_0^{2\pi}\left(a(1-\cos\theta)\right)^2\mathrm{d}\theta = \left[\frac{1}{2}a^2\left(\frac{3}{2}\theta-2\sin\theta+\frac{1}{4}\sin(2\theta)\right)\right]_0^{2\pi}$

$= \left(\frac{1}{2}a^2\left(\frac{3}{2}\times 2\pi-0+0\right)\right)-\left(\frac{1}{2}a^2(0-0+0)\right)$

$= \frac{3}{2}a^2\pi$ units^2

2) d)

$\frac{1}{2}\int \left(a(1-\sin\theta)\right)^2 \mathrm{d}\theta = \frac{1}{2}a^2\int (1-2\sin\theta+\sin^2\theta)\mathrm{d}\theta$

Since $\cos(2A) = 1-2\sin^2 A$ then $\sin^2 A = \frac{1}{2}-\frac{1}{2}\cos(2A)$

$\frac{1}{2}a^2\int (1-2\sin\theta+\sin^2\theta)\mathrm{d}\theta = \frac{1}{2}a^2\int \left(1-2\sin\theta+\frac{1}{2}-\frac{1}{2}\cos(2\theta)\right)\mathrm{d}\theta$

$= \frac{1}{2}a^2\int \left(\frac{3}{2}-2\sin\theta-\frac{1}{2}\cos(2\theta)\right)\mathrm{d}\theta$

$= \frac{1}{2}a^2\left(\frac{3}{2}\theta-2\cos\theta-\frac{1}{4}\sin(2\theta)\right)(+c)$

So that

$\frac{1}{2}\int_0^{2\pi}\left(a(1-\sin\theta)\right)^2\mathrm{d}\theta = \left[\frac{1}{2}a^2\left(\frac{3}{2}\theta-2\cos\theta-\frac{1}{4}\sin(2\theta)\right)\right]_0^{2\pi}$

$= \left(\frac{1}{2}a^2\left(\frac{3}{2}\times 2\pi-2-0\right)\right)-\left(\frac{1}{2}a^2(0-2-0)\right)$

$= \frac{3}{2}a^2\pi$ units^2

3) a)

Since $\cos(2A) = 2\cos^2 A-1$ then $\cos^2 A = \frac{1}{2}+\frac{1}{2}\cos(2A)$

$\frac{1}{2}\int (a\cos\theta)^2\mathrm{d}\theta = \frac{1}{2}a^2\int \left(\frac{1}{2}+\frac{1}{2}\cos(2\theta)\right)\mathrm{d}\theta$

$= \frac{1}{2}a^2\left(\frac{1}{2}\theta+\frac{1}{4}\sin(2\theta)\right)(+c)$

So that

$\frac{1}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(a\cos\theta)^2\mathrm{d}\theta = \left[\frac{1}{2}a^2\left(\frac{1}{2}\theta+\frac{1}{4}\sin(2\theta)\right)\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$

$= \left(\frac{1}{2}a^2\left(\frac{\pi}{4}+0\right)\right)-\left(\frac{1}{2}a^2\left(-\frac{\pi}{4}+0\right)\right)$

$= \frac{1}{4}a^2\pi$ units^2

3) b)

Since $\cos(2A) = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$

$\frac{1}{2}\int (a\sin\theta)^2 \mathrm{d}\theta = \frac{1}{2}a^2 \int \left(\frac{1}{2} - \frac{1}{2}\cos(2\theta)\right) \mathrm{d}\theta$

$= \frac{1}{2}a^2\left(\frac{1}{2}\theta - \frac{1}{4}\sin(2\theta)\right)(+c)$

So that

$\frac{1}{2}\int_0^{\pi} (a\sin\theta)^2 \mathrm{d}\theta = \left[\frac{1}{2}a^2\left(\frac{1}{2}\theta - \frac{1}{4}\sin(2\theta)\right)\right]_0^{\pi}$

$= \left(\frac{1}{2}a^2\left(\frac{1}{2}\pi - 0\right)\right) - \left(\frac{1}{2}a^2(0-0)\right)$

$= \frac{1}{4}a^2\pi$ units^2

3) c)

$\frac{1}{2}\int a^2 \mathrm{d}\theta = \frac{1}{2}a^2 \int 1 \mathrm{d}\theta$

$= \frac{1}{2}a^2\theta(+c)$

So that

$\frac{1}{2}\int a^2 \mathrm{d}\theta = \left[\frac{1}{2}a^2\theta\right]_0^{2\pi}$

$= (a^2 \times \pi) - (0)$

$= a^2\pi$ units^2

4) a)

$\frac{1}{2}\int_0^{\pi} \left(a\mathrm{e}^{\theta}\right)^2 \mathrm{d}\theta = \frac{1}{2}a^2 \int_0^{\pi} \mathrm{e}^{2\theta} \mathrm{d}\theta$

$= \left[\frac{1}{4}a^2\mathrm{e}^{2\theta}\right]_0^{\pi}$

$= \left(\frac{1}{4}a^2\mathrm{e}^{2\pi}\right) - \left(\frac{1}{4}a^2\right)$

$= \frac{1}{4}a^2(\mathrm{e}^{2\pi} - 1)$ units^2

4) b)

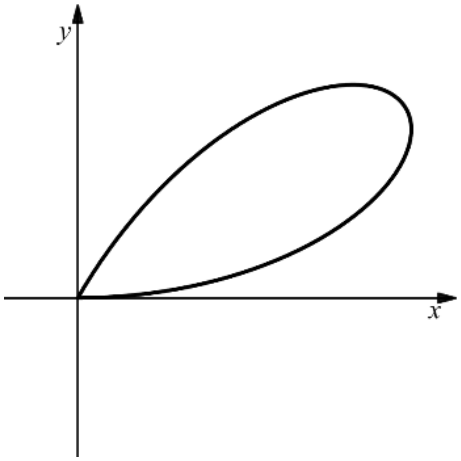
$\frac{1}{2}\int_0^{\pi}(a\theta)^2 \mathrm{d}\theta = \frac{1}{2}a^2\int_0^{\pi}\theta^2\,\mathrm{d}\theta$

$= \left[\frac{1}{6}a^2\theta^3\right]_0^{\pi}$

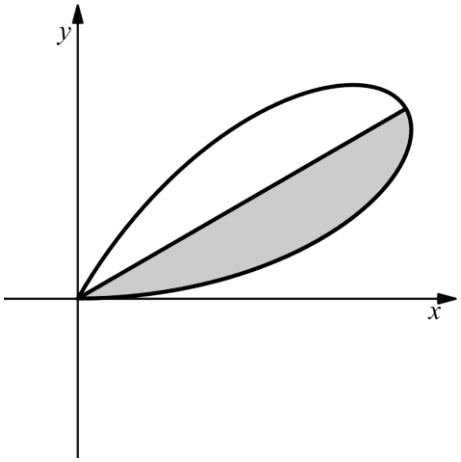
$= \left(\frac{1}{6}a^2\pi^3\right) - (0)$

$= \frac{1}{6}a^2\pi^3$ units^2

5) a)



5) b)



5) c)

$\frac{1}{2}\int_0^{\frac{\pi}{6}}(2\cos 3\theta)^2\,\mathrm{d}\theta = 2\int_0^{\frac{\pi}{6}}\cos^2(3\theta)\,\mathrm{d}\theta$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

$2\int_0^{\frac{\pi}{6}}\cos^2(3\theta)\,\mathrm{d}\theta = 2\int_0^{\frac{\pi}{6}}\left(\frac{1}{2} + \frac{1}{2}\cos(6\theta)\right)\mathrm{d}\theta = \int_0^{\frac{\pi}{6}}(1 + \cos(6\theta))\,\mathrm{d}\theta$

$= \left[\theta + \frac{1}{6}\sin(6\theta)\right]_0^{\frac{\pi}{6}}$

$= \left(\frac{\pi}{6} + 0\right) - (0 + 0)$

$= \frac{\pi}{6}$ units^2

6) a)

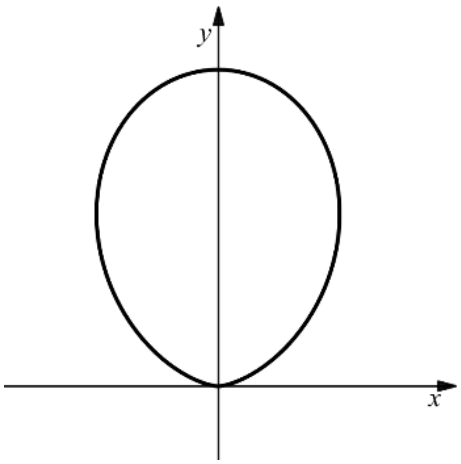
When $\theta = 0$, $r = 0$ and when $\theta = \pi$, $r = 0$ so it starts and ends at the pole.

For $0 < \theta < \pi$:

$\sin\theta$ increases to a maximum before decreasing, so $\sin^2\theta$ increases to a maximum before decreasing

$\therefore$ The curve $r = a\sin^2\theta$ for $0 \leq \theta \leq \pi$ for positive constant a consists of a single closed loop through the pole.

6) b)



6) c)

$\frac{1}{2}\int_0^{\pi}(a\sin^2\theta)^2\,\mathrm{d}\theta = \frac{1}{2}a^2\int_0^{\pi}\sin^4\theta\,\mathrm{d}\theta$

Since $\cos(2A) = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$

So $\sin^4 A = \left(\frac{1}{2} - \frac{1}{2}\cos(2A)\right)\left(\frac{1}{2} - \frac{1}{2}\cos(2A)\right) = \frac{1}{4} - \frac{1}{2}\cos(2A) + \frac{1}{4}\cos^2(2A)$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

So $\sin^4 A = \frac{1}{4} - \frac{1}{2}\cos(2A) + \frac{1}{4}\left(\frac{1}{2} + \frac{1}{2}\cos(4A)\right) = \frac{3}{8} - \frac{1}{2}\cos(2A) + \frac{1}{8}\cos(4A)$

$\frac{1}{2}a^2\int_0^{\pi}\sin^4\theta\,\mathrm{d}\theta = \frac{1}{2}a^2\int_0^{\pi}\left(\frac{3}{8} - \frac{1}{2}\cos(2\theta) + \frac{1}{8}\cos(4\theta)\right)\mathrm{d}\theta$

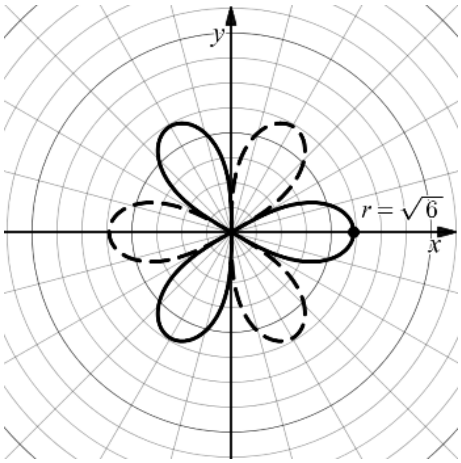
$= \left[\frac{1}{2}a^2\left(\frac{3}{8}\theta - \frac{1}{4}\sin(2\theta) + \frac{1}{32}\sin(4\theta)\right)\right]_0^{\pi}$

$= \left(\frac{1}{2}a^2\left(\frac{3}{8}\pi - 0 + 0\right)\right) - \left(\frac{1}{2}a^2(0 - 0 + 0)\right)$

$= \frac{3}{16}\pi a^2$ units^2

7) a)

Since $r^2 = 6\cos 3\theta$ then consider $r = \pm\sqrt{6\cos 3\theta}$ and use dashed lines for negative values of r



7) b)

The student has found the sum of both the positive contributions and the negative contributions, so they have cancelled out to form to the answer zero.

7) c)

$\frac{1}{2}\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 6\cos 3\theta \, \mathrm{d}\theta = \left[\frac{1}{2}(2\sin 3\theta)\right]_{-\frac{\pi}{6}}^{\frac{\pi}{6}}$

$= \left(\frac{1}{2}\times 2\right) - \left(\frac{1}{2}\times -2\right)$

$= 2$ units^2

7) d)

Total area is 12 units^2

8) a)

$\sin 2x = 0$ at $x = 0$ and $x = \frac{\pi}{2}$

$\frac{1}{2}\int_{0}^{\frac{\pi}{2}} a\sin 2\theta \, \mathrm{d}\theta = \left[-\frac{1}{4}a\cos 2\theta\right]_{0}^{\frac{\pi}{2}}$

$= \left(\frac{1}{4}a\right) - \left(-\frac{1}{4}a\right)$

$= \frac{1}{2}a$

So two loops gives: $2\times\frac{1}{2}a = \ a$ units^2

8) b)

$\cos 2x = 0$ at $x = -\frac{\pi}{4}$ and $x = \frac{\pi}{4}$

$\frac{1}{2}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} a\cos 2\theta \, \mathrm{d}\theta = \left[\frac{1}{4}a\sin 2\theta\right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}}$

$= \left(\frac{1}{4}a\right) - \left(-\frac{1}{4}a\right)$

$= \frac{1}{2}a$

So two loops gives: $2\times\frac{1}{2}a = \ a$ units^2

8) c)

$\sin 3x = 0$ at $x = 0$ and $x = \frac{\pi}{3}$

$\frac{1}{2}\int_0^{\frac{\pi}{3}} a \sin 3\theta \, \mathrm{d}\theta = \left[-\frac{1}{6}a\cos 3\theta\right]_0^{\frac{\pi}{3}}$

$= \left(\frac{1}{6}a\right) - \left(-\frac{1}{6}a\right)$

$= \frac{1}{3}a$

So six loops gives: $6 \times \frac{1}{3}a = 2a$ units^2

8) d)

$\cos 3x = 0$ at $x = -\frac{\pi}{6}$ and $x = \frac{\pi}{6}$

$\frac{1}{2}\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} a \cos 3\theta \, \mathrm{d}\theta = \left[\frac{1}{6}a\sin 3\theta\right]_{-\frac{\pi}{6}}^{\frac{\pi}{6}}$

$= \left(\frac{1}{6}a\right) - \left(-\frac{1}{6}a\right)$

$= \frac{1}{3}a$

So six loops gives: $6 \times \frac{1}{3}a = 2a$ units^2

9) a)

$\cos 3x = 0$ at $x = -\frac{\pi}{6}$ and $x = \frac{\pi}{6}$

$\frac{1}{2}\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} (2\cos(3\theta))^2 \mathrm{d}\theta = 2\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos^2(3\theta) \, \mathrm{d}\theta$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

$2\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos^2(3\theta) \, \mathrm{d}\theta = \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} (1 + 1\cos(6\theta)) \mathrm{d}\theta$

$= \left[\theta + \frac{1}{6}\sin(6\theta)\right]_{-\frac{\pi}{6}}^{\frac{\pi}{6}}$

$= \left(\frac{\pi}{6} + 0\right) - \left(-\frac{\pi}{6} + 0\right)$

$= \frac{\pi}{3}$

So three loops gives: $3 \times \frac{\pi}{3} = \pi$ units^2

9) b)

$\sin 2x = 0$ at $x = 0$ and $x = \frac{\pi}{2}$

$\frac{1}{2}\int_0^{\frac{\pi}{2}}(3\sin(2\theta))^2 \mathrm{d}\theta = \frac{9}{2}\int_0^{\frac{\pi}{2}}\sin^2(2\theta)\,\mathrm{d}\theta$

Since $\cos(2A) = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$

$\frac{9}{2}\int_0^{\frac{\pi}{2}}\sin^2(2\theta)\,\mathrm{d}\theta = \frac{9}{4}\int_0^{\frac{\pi}{2}}(1 - \cos(4\theta))\mathrm{d}\theta$

$= \left[\frac{9}{4}\left(\theta - \frac{1}{4}\sin(4\theta)\right)\right]_0^{\frac{\pi}{2}}$

$= \left(\frac{9}{4}\left(\frac{\pi}{2} - 0\right)\right) - (0 - 0)$

$= \frac{9\pi}{8}$

So four loops gives: $4 \times \frac{9\pi}{8} = \frac{9\pi}{2}$ units^2

9) c)

$\cos 4x = 0$ at $x = -\frac{\pi}{8}$ and $x = \frac{\pi}{8}$

$\frac{1}{2}\int_{-\frac{\pi}{8}}^{\frac{\pi}{8}}(a\cos(4\theta))^2 \mathrm{d}\theta = \frac{1}{2}a^2\int_{-\frac{\pi}{8}}^{\frac{\pi}{8}}\cos^2(4\theta)\,\mathrm{d}\theta$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

$\frac{1}{2}a^2\int_{-\frac{\pi}{8}}^{\frac{\pi}{8}}\cos^2(4\theta)\,\mathrm{d}\theta = \frac{1}{4}a^2\int_{-\frac{\pi}{8}}^{\frac{\pi}{8}}(1 + 1\cos(8\theta))\mathrm{d}\theta$

$= \left[\frac{1}{4}a^2\left(\theta + \frac{1}{8}\sin(8\theta)\right)\right]_{-\frac{\pi}{8}}^{\frac{\pi}{8}}$

$= \left(\frac{1}{4}a^2\left(\frac{\pi}{8} + 0\right)\right) - \left(\frac{1}{4}a^2\left(-\frac{\pi}{8} + 0\right)\right)$

$= \frac{1}{16}\pi a^2$

So eight loops gives: $8 \times \frac{1}{16}\pi a^2 = \frac{1}{2}\pi a^2$ units^2

9) d)

$\sin 5x = 0$ at $x = 0$ and $x = \frac{\pi}{5}$

$\frac{1}{2}\int_0^{\frac{\pi}{5}}(a\sin(5\theta))^2 \mathrm{d}\theta = \frac{1}{2}a^2\int_0^{\frac{\pi}{5}}\sin^2(5\theta)\,\mathrm{d}\theta$

Since $\cos(2A) = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$

$\frac{1}{2}a^2\int_0^{\frac{\pi}{5}}\sin^2(5\theta)\,\mathrm{d}\theta = \frac{1}{4}a^2\int_0^{\frac{\pi}{5}}(1-\cos(10\theta))\mathrm{d}\theta$

$= \left[\frac{1}{4}a^2\left(\theta - \frac{1}{10}\sin(10\theta)\right)\right]_0^{\frac{\pi}{5}}$

$= \left(\frac{1}{4}a^2\left(\frac{\pi}{5} - 0\right)\right) - (0-0)$

$= \frac{1}{20}\pi a^2$

So five loops gives: $5 \times \frac{1}{20}\pi a^2 = \frac{1}{4}\pi a^2$ units^2

10)

$\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{11\pi}{6}}\left(\frac{1}{\sqrt{\theta}}\right)^2 \mathrm{d}\theta = \frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{11\pi}{6}}\frac{1}{\theta}\mathrm{d}\theta = \left[\frac{1}{2}\ln\theta\right]_{\frac{\pi}{6}}^{\frac{11\pi}{6}}$

$= \left(\frac{1}{2}\ln\frac{11\pi}{6}\right) - \left(\frac{1}{2}\ln\frac{\pi}{6}\right)$

$= \frac{1}{2}\ln\left(\frac{\frac{11\pi}{6}}{\frac{\pi}{6}}\right)$

$= \ln\sqrt{11}$ units^2

11) a)

$0 = 1 + 2\cos\theta \Rightarrow \theta = \frac{2\pi}{3}$ and $\theta = \frac{4\pi}{3}$

So negative for $\frac{2\pi}{3} < \theta < \frac{4\pi}{3}$

$\frac{1}{2}\int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}}(1 + 2\cos\theta)^2 \mathrm{d}\theta = \frac{1}{2}\int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}}(1 + 4\cos\theta + 4\cos^2\theta)\mathrm{d}\theta$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

$\frac{1}{2}\int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}}(1 + 4\cos\theta + 4\cos^2\theta)\mathrm{d}\theta = \frac{1}{2}\int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}}(3 + 4\cos\theta + 2\cos(2\theta))\mathrm{d}\theta$

$= \left[\frac{3}{2}\theta + 2\sin\theta + \frac{1}{2}\sin(2\theta)\right]_{\frac{2\pi}{3}}^{\frac{4\pi}{3}}$

$= \left(\frac{3}{2} \times \frac{4\pi}{3} - \sqrt{3} + \frac{\sqrt{3}}{4}\right) - \left(\frac{3}{2} \times \frac{2\pi}{3} + \sqrt{3} - \frac{\sqrt{3}}{4}\right)$

$= \left(\pi - \frac{3\sqrt{3}}{2}\right)$ units^2

11) b)

$0 = 2 - 4\sin\theta \Rightarrow \theta = \frac{\pi}{6}$ and $\theta = \frac{5\pi}{6}$

So negative for $\frac{\pi}{6} < \theta < \frac{5\pi}{6}$

$\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}}(2 - 4\sin\theta)^2 \mathrm{d}\theta = \frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}}(4 - 16\sin\theta + 16\sin^2\theta)\mathrm{d}\theta$

Since $\cos(2A) = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$

$\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}}(4 - 16\sin\theta + 16\sin^2\theta)\mathrm{d}\theta = \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}}(6 - 8\sin\theta - 4\cos(2\theta))\mathrm{d}\theta$

$= [6\theta + 8\cos\theta - 2\sin(2\theta)]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$

$= \left(6 \times \frac{5\pi}{6} - 4\sqrt{3} + \sqrt{3}\right) - \left(6 \times \frac{\pi}{6} + 4\sqrt{3} - \sqrt{3}\right)$

$= \left(4\pi - 6\sqrt{3}\right)$ units^2

12)

$\frac{1}{2}\int_0^{2\pi}\big(a(p+q\sin\theta)\big)^2\mathrm{d}\theta = \frac{1}{2}a^2\int_0^{2\pi}(p^2+2pq\sin\theta+q^2\sin\theta)\mathrm{d}\theta$

Since $\cos(2A) = 1-2\sin^2 A$ then $\sin^2 A = \frac{1}{2}-\frac{1}{2}\cos(2A)$

$\frac{1}{2}a^2\int_0^{2\pi}(p^2+2pq\sin\theta+q^2\sin\theta)\mathrm{d}\theta$

$= \frac{1}{2}a^2\int_0^{2\pi}\left(p^2+2pq\sin\theta+\frac{1}{2}q^2-\frac{1}{2}q^2\cos(2\theta)\right)\mathrm{d}\theta$

$= \left[\frac{1}{2}a^2\left(\left(p^2+\frac{1}{2}q^2\right)\theta-2pq\cos\theta-\frac{1}{4}q^2\sin(2\theta)\right)\right]_0^{2\pi}$

$= \left(\frac{1}{2}a^2\left(\left(p^2+\frac{1}{2}q^2\right)2\pi-2pq-0\right)\right)-\left(\frac{1}{2}a^2(0-2pq-0)\right)$

$= \left(\frac{1}{2}a^2p^2+\frac{1}{4}a^2q^2\right)2\pi-a^2pq+a^2pq$

$= 2\pi a^2\left(\frac{1}{2}p^2+\frac{1}{4}q^2\right)$

$= \frac{1}{2}\pi a^2(2p^2+q^2)$ units^2 as req.

13) a)

The curve is symmetrical across the initial line so that the area is $1:1$

13) b)

$\frac{1}{2}\int_0^{\pi}\left(2(1+\sin\theta)\right)^2\mathrm{d}\theta = 2\int_0^{\pi}(1+2\sin\theta+\sin\theta)\mathrm{d}\theta$

Since $\cos(2A) = 1-2\sin^2 A$ then $\sin^2 A = \frac{1}{2}-\frac{1}{2}\cos(2A)$

$2\int_0^{\pi}(1+2\sin\theta+\sin\theta)\mathrm{d}\theta$

$= 2\int_0^{\pi}\left(\frac{3}{2}+2\sin\theta-\frac{1}{2}\cos(2\theta)\right)\mathrm{d}\theta$

$= \left[2\left(\frac{3}{2}\theta-2\cos\theta-\frac{1}{4}\sin(2\theta)\right)\right]_0^{\pi}$

$= \left(2\left(\frac{3}{2}\times\pi+2-0\right)\right)-\left(2(0-2-0)\right)$

$= 8+3\pi$

$\frac{1}{2}\int_{\pi}^{2\pi}\left(2(1+\sin\theta)\right)^2\mathrm{d}\theta = 2\int_{\pi}^{2\pi}(1+2\sin\theta+\sin^2\theta)\mathrm{d}\theta$

Since $\cos(2A) = 1-2\sin^2 A$ then $\sin^2 A = \frac{1}{2}-\frac{1}{2}\cos(2A)$

$2\int_{\pi}^{2\pi}(1+2\sin\theta+\sin 2\theta)\mathrm{d}\theta$

$= 2\int_0^{2\pi}\left(\frac{3}{2}+2\sin\theta-\frac{1}{2}\cos(2\theta)\right)\mathrm{d}\theta$

$= \left[2\left(\frac{3}{2}\theta-2\cos\theta-\frac{1}{4}\sin(2\theta)\right)\right]_{\pi}^{2\pi}$

$= \left(2\left(\frac{3}{2}\times 2\pi-2-0\right)\right)-\left(2\left(\frac{3}{2}\times\pi+2-0\right)\right)$

$= -8+3\pi$

So the ratio is $(8+3\pi):(-8+3\pi)$

14) a)

$\frac{1}{2}\int_0^{2\pi}\left(\frac{2}{5}(4+\sin 4\theta)\right)^2\mathrm{d}\theta = \frac{2}{25}\int_0^{2\pi}(16+8\sin 4\theta+\sin^2 4\theta)\mathrm{d}\theta$

Since $\cos(2A) = 1-2\sin^2 A$ then $\sin^2 A = \frac{1}{2}-\frac{1}{2}\cos(2A)$

$\frac{2}{25}\int_0^{2\pi}(16+8\sin 4\theta+\sin^2 4\theta)\mathrm{d}\theta = \frac{2}{25}\int_0^{2\pi}\left(\frac{33}{2}+8\sin 4\theta-\frac{1}{2}\cos(8\theta)\right)\mathrm{d}\theta$

$= \left[\frac{2}{25}\left(\frac{33}{2}\theta-2\cos 4\theta-\frac{1}{16}\sin(8\theta)\right)\right]_0^{2\pi}$

$= \left(\frac{2}{25}\left(\frac{33}{2}\times 2\pi-2-0\right)\right)-\left(\frac{2}{25}(0-2-0)\right)$

$= \frac{66}{25}\pi$

$0.8\times\frac{66}{25}\pi = \frac{264}{125}\pi\ \mathrm{m}^3$

14) b)

All the water stays in the pond, no evaporation or overflow. The pond is filled exactly to the top. That the pond exactly matches the curve.

15) a)

$\frac{1}{2}\int_0^{2\pi}(5+\cos 4\theta)^2 \mathrm{d}\theta = \frac{1}{2}\int_0^{2\pi}(25+10\cos 4\theta+\cos^2 4\theta)\mathrm{d}\theta$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2}+\frac{1}{2}\cos(2A)$

$\frac{1}{2}\int_0^{2\pi}(25+10\cos 4\theta+\cos^2 4\theta)\mathrm{d}\theta = \frac{1}{2}\int_0^{2\pi}\left(\frac{51}{2}+10\cos 4\theta+\frac{1}{2}\cos 8\theta\right)\mathrm{d}\theta$

$= \left[\frac{51}{4}\theta+\frac{5}{4}\sin 4\theta+\frac{1}{32}\sin(8\theta)\right]_0^{2\pi}$

$= \left(\frac{51}{4}\times 2\pi+0+0\right)-(0+0+0)$

$= \frac{51}{2}\pi$

$0.4\times\frac{51}{2}\pi = \frac{51}{5}\pi$ m^3

40 litres per minute is equivalent to 0.04 m^3 per minute

So the time is $\frac{\frac{51}{5}\pi}{0.04} = 801$ minutes

15) b)

The pool is assumed to have vertical sides at every point. The pool has uniform depth. The pool is filled exactly to the top (not less than the full amount or with over-spill). The flow rate of the water is constant as claimed.

16) a)

$\frac{1}{2}\int_0^{\frac{\pi}{4}}\left(a\mathrm{e}^{k\theta}\right)^2\mathrm{d}\theta = \frac{1}{2}a^2\int_0^{\frac{\pi}{4}}\mathrm{e}^{2k\theta}\mathrm{d}\theta = \left[\frac{1}{2}a^2\left(\frac{1}{2k}\mathrm{e}^{2k\theta}\right)\right]_0^{\frac{\pi}{4}}$

$= \frac{1}{4k}a^2\left(\left(\mathrm{e}^{\frac{2\pi}{4}k}\right)-\left(\mathrm{e}^0\right)\right) = \frac{1}{4k}a^2\left(\mathrm{e}^{\frac{\pi}{2}k}-1\right)$ units^2

16) b)

$\frac{1}{2}\int_{2\pi}^{\frac{9\pi}{4}}\left(a\mathrm{e}^{k\theta}\right)^2\mathrm{d}\theta = \frac{1}{2}a^2\int_{2\pi}^{\frac{9\pi}{4}}\mathrm{e}^{2k\theta}\mathrm{d}\theta = \left[\frac{1}{2}a^2\left(\frac{1}{2k}\mathrm{e}^{2k\theta}\right)\right]_{2\pi}^{\frac{9\pi}{4}}$

$= \frac{1}{4k}a^2\left(\left(\mathrm{e}^{\frac{9\pi}{2}k}\right)-\left(\mathrm{e}^{4\pi k}\right)\right) = \frac{1}{4k}a^2\left(\mathrm{e}^{\frac{9\pi}{2}k}-\mathrm{e}^{4\pi k}\right) = \frac{1}{4k}a^2\mathrm{e}^{4\pi k}\left(\mathrm{e}^{\frac{\pi}{2}k}-1\right)$ units^2

16) c)

$\frac{1}{2}\int_{4\pi}^{\frac{17\pi}{4}}\left(a\mathrm{e}^{k\theta}\right)^2\mathrm{d}\theta = \frac{1}{2}a^2\int_{4\pi}^{\frac{17\pi}{4}}\mathrm{e}^{2k\theta}\mathrm{d}\theta = \left[\frac{1}{2}a^2\left(\frac{1}{2k}\mathrm{e}^{2k\theta}\right)\right]_{4\pi}^{\frac{17\pi}{4}}$

$= \frac{1}{4k}a^2\left(\left(\mathrm{e}^{\frac{17\pi}{2}k}\right) - \left(\mathrm{e}^{8\pi k}\right)\right) = \frac{1}{4k}a^2\left(\mathrm{e}^{\frac{17\pi}{2}k} - \mathrm{e}^{8\pi k}\right) = \frac{1}{4k}a^2\mathrm{e}^{8\pi k}\left(\mathrm{e}^{\frac{\pi}{2}k} - 1\right)$ units^2

16) d)

For $n = 0$: $\frac{1}{4k}a^2\left(\mathrm{e}^{\frac{\pi}{2}k} - 1\right)$

For $n = 1$: $\frac{1}{4k}a^2\mathrm{e}^{4\pi k}\left(\mathrm{e}^{\frac{\pi}{2}k} - 1\right)$

For $n = 2$: $\frac{1}{4k}a^2\mathrm{e}^{8\pi k}\left(\mathrm{e}^{\frac{\pi}{2}k} - 1\right)$

Which is a geometric sequence with first term $\frac{1}{4k}a^2\left(\mathrm{e}^{\frac{\pi}{2}k} - 1\right)$ and common ratio $\mathrm{e}^{4\pi k}$

17)

$\frac{1}{2}\int_0^{\pi}(1+\sinh\theta)^2\mathrm{d}\theta = \frac{1}{2}\int_0^{\pi}(1+2\sinh\theta+\sinh^2\theta)\mathrm{d}\theta$

Since $\cosh(2A) = 1 + 2\sinh^2 A$ then $\sinh^2 A = \frac{1}{2}\cosh(2A) - \frac{1}{2}$

$\frac{1}{2}\int_0^{\pi}(1+2\sinh\theta+\sinh^2\theta)\mathrm{d}\theta = \frac{1}{2}\int_0^{\pi}\left(1+2\sinh\theta+\left(\frac{1}{2}\cosh(2\theta)-\frac{1}{2}\right)\right)\mathrm{d}\theta$

$= \frac{1}{2}\int_0^{\pi}\left(\frac{1}{2}+2\sinh\theta+\frac{1}{2}\cosh(2\theta)\right)\mathrm{d}\theta$

$= \left[\frac{1}{4}\theta+\cosh\theta+\frac{1}{8}\sinh(2\theta)\right]_0^{\pi}$

$= \left(\frac{1}{4}\pi+\cosh\pi+\frac{1}{8}\sinh(2\pi)\right) - (0+1+0)$

$= \frac{1}{4}\pi - 1 + \cosh\pi + \frac{1}{8}\sinh(2\pi)$

$= 44.8$ units^2 (3 SF)

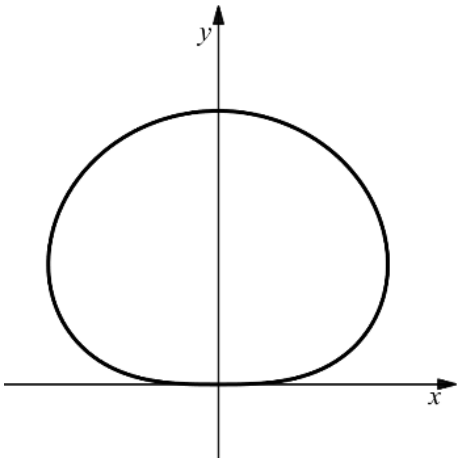
18) a)

Since $y = \sin x$ is positive for $0 < x < \pi$ then $k = \pi$

18) b)

Greatest value of r is when $\sin\theta = 1$ so $r = \sqrt{3}$

18) c)



18) d)

$\frac{1}{2}\int_0^{\pi}\left(\sqrt{3\sin\theta}\right)^2 \mathrm{d}\theta = \frac{1}{2}\int_0^{\pi} 3\sin\theta\,\mathrm{d}\theta = \left[-\frac{3}{2}\cos\theta\right]_0^{\pi} = \left(-\frac{3}{2}\cos\pi\right) - \left(-\frac{3}{2}\cos 0\right) = 3$ units^2

19) a)

Since $y = \sin 2x$ is positive for $0 < x < \frac{\pi}{2}$ then $k = \frac{\pi}{2}$

19) b)

$\frac{1}{2}\int_0^{\frac{\pi}{2}}\left(\sqrt{16\sin 2\theta}\right)^2 \mathrm{d}\theta = 8\int_0^{\frac{\pi}{2}}\sin 2\theta\,\mathrm{d}\theta = [-4\cos 2\theta]_0^{\frac{\pi}{2}} = (-4\cos\pi) - (-4\cos 0)$

$= 8$ units^2

19) c)

r is a maximum when $\sin 2\theta = 1$ so $r = 4$

This occurs when $\theta = \frac{\pi}{4}$ since $\sqrt{16\sin\left(2\times\frac{\pi}{4}\right)} = 4$

So $\left(4, \frac{\pi}{4}\right)$

19) d)

Cartesian coordinates of P are $\left(4\cos\frac{\pi}{4}, 4\sin\frac{\pi}{4}\right) = \left(2\sqrt{2}, 2\sqrt{2}\right)$

So $\begin{pmatrix}1 & 3\\ 0 & 1\end{pmatrix}\begin{pmatrix}2\sqrt{2}\\ 2\sqrt{2}\end{pmatrix} = \begin{pmatrix}8\sqrt{2}\\ 2\sqrt{2}\end{pmatrix}$

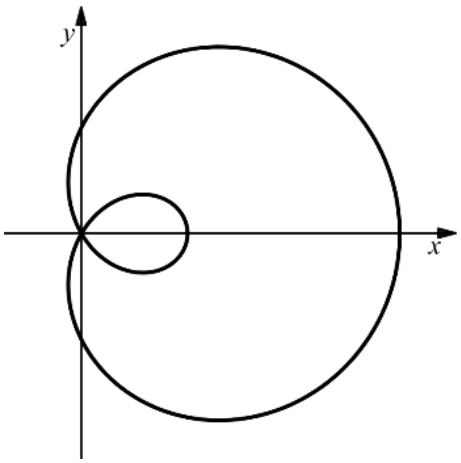
So the image of P is at $\left(8\sqrt{2}, 2\sqrt{2}\right)$

That is $r\cos\theta = 8\sqrt{2}$ and $r\sin\theta = 2\sqrt{2} \therefore \tan\theta = \frac{2\sqrt{2}}{8\sqrt{2}} = \frac{1}{4}$

19) e)

$\det\mathbf{M} = 1 - 0 = 1$ so the area remains the same: 8 units^2

20) a)



20) b)

Inside loop is when $\frac{2\pi}{3} \leq \theta < \frac{4\pi}{3}$

So the area wanted is:

$$\frac{1}{2}\int_0^{\frac{2\pi}{3}}(2+4\cos\theta)^2\mathrm{d}\theta + \frac{1}{2}\int_{\frac{4\pi}{3}}^{2\pi}(2+4\cos\theta)^2\mathrm{d}\theta - \frac{1}{2}\int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}}(2+4\cos\theta)^2\mathrm{d}\theta$$

Find the integral:

$$\frac{1}{2}\int(2+4\cos\theta)^2\,\mathrm{d}\theta = \int(2+8\cos\theta+8\cos^2\theta)\,\mathrm{d}\theta$$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

$$\int(2+8\cos\theta+8\cos^2\theta)\,\mathrm{d}\theta = \int(6+8\cos\theta+4\cos(2\theta))\,\mathrm{d}\theta$$

$$= 6\theta + 8\sin\theta + 2\sin 2\theta\ (+c)$$

So each area is:

$$\frac{1}{2}\int_0^{\frac{2\pi}{3}}(2+4\cos\theta)^2\mathrm{d}\theta = [6\theta+8\sin\theta+2\sin 2\theta]_0^{\frac{2\pi}{3}}$$

$$= \left(4\pi + 4\sqrt{3} - \sqrt{3}\right) - (0+0+0) = 4\pi + 3\sqrt{3}$$

By symmetry: $\frac{1}{2}\int_{\frac{4\pi}{3}}^{2\pi}(2+4\cos\theta)^2\mathrm{d}\theta = 4\pi + 3\sqrt{3}$

$$\frac{1}{2}\int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}}(2+4\cos\theta)^2\mathrm{d}\theta = [6\theta+8\sin\theta+2\sin 2\theta]_{\frac{2\pi}{3}}^{\frac{4\pi}{3}}$$

$$= \left(8\pi - 4\sqrt{3} + \sqrt{3}\right) - \left(4\pi + 4\sqrt{3} - \sqrt{3}\right) = 4\pi - 6\sqrt{3}$$

Therefore

$$\frac{1}{2}\int_0^{\frac{2\pi}{3}}(2+4\cos\theta)^2\mathrm{d}\theta + \frac{1}{2}\int_{\frac{4\pi}{3}}^{2\pi}(2+4\cos\theta)^2\mathrm{d}\theta - \frac{1}{2}\int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}}(2+4\cos\theta)^2\mathrm{d}\theta$$

$$= 4\pi + 3\sqrt{3} + 4\pi + 3\sqrt{3} - \left(4\pi - 6\sqrt{3}\right)$$

$= 4\pi + 12\sqrt{3}$ units^2 as req.

21)

$y = 2 - \sin\left(\frac{\theta}{4}\right)$ is a graph symmetrical about $\theta = 2\pi$ with maximum of $r = 3$ when $\theta = 0$ and $\theta = 4\pi$ and a minimum of $r = 1$ when $r = 2\pi$

So the shaded area is

$$\frac{1}{2}\int_0^{\pi}\left(2-\sin\left(\frac{\theta}{4}\right)\right)^2\mathrm{d}\theta - \frac{1}{2}\int_{\pi}^{3\pi}\left(2-\sin\left(\frac{\theta}{4}\right)\right)^2\mathrm{d}\theta + \frac{1}{2}\int_{3\pi}^{4\pi}\left(2-\sin\left(\frac{\theta}{4}\right)\right)^2\mathrm{d}\theta$$

Find the integral:

$$\frac{1}{2}\int\left(2-\sin\left(\frac{\theta}{4}\right)\right)^2\mathrm{d}\theta = \frac{1}{2}\int\left(4-4\sin\left(\frac{\theta}{4}\right)+\sin^2\left(\frac{\theta}{4}\right)\right)\mathrm{d}\theta$$

Since $\cos(2A) = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$

$\frac{1}{2}\int \left(4 - 4\sin\left(\frac{\theta}{4}\right) + \sin^2\left(\frac{\theta}{4}\right)\right) \mathrm{d}\theta = \int \left(\frac{9}{4} - 2\sin\left(\frac{\theta}{4}\right) - \frac{1}{4}\cos\left(\frac{\theta}{2}\right)\right) \mathrm{d}\theta$

$= \frac{9}{4}\theta + 8\cos\left(\frac{\theta}{4}\right) - \frac{1}{2}\sin\left(\frac{\theta}{2}\right) (+c)$

So each area is:

$\frac{1}{2}\int_0^{\pi} \left(2 - \sin\left(\frac{\theta}{4}\right)\right)^2 \mathrm{d}\theta = \left[\frac{9}{4}\theta + 8\cos\left(\frac{\theta}{4}\right) - \frac{1}{2}\sin\left(\frac{\theta}{2}\right)\right]_0^{\pi}$

$= \left(\frac{9}{4}\pi + 4\sqrt{2} - \frac{1}{2}\right) - (0 + 8 - 0) = -\frac{17}{2} + 4\sqrt{2} + \frac{9\pi}{4}$

$\frac{1}{2}\int_{\pi}^{3\pi} \left(2 - \sin\left(\frac{\theta}{4}\right)\right)^2 \mathrm{d}\theta = \left[\frac{9}{4}\theta + 8\cos\left(\frac{\theta}{4}\right) - \frac{1}{2}\sin\left(\frac{\theta}{2}\right)\right]_{\pi}^{3\pi}$

$= \left(\frac{9}{4} \times 3\pi - 4\sqrt{2} + \frac{1}{2}\right) - \left(\frac{9}{4} \times \pi + 4\sqrt{2} - \frac{1}{2}\right) = 1 - 8\sqrt{2} + \frac{9\pi}{2}$

By symmetry: $\frac{1}{2}\int_{3\pi}^{4\pi} \left(2 - \sin\left(\frac{\theta}{4}\right)\right)^2 \mathrm{d}\theta = -\frac{17}{2} + 4\sqrt{2} + \frac{9\pi}{4}$

So

$\frac{1}{2}\int_0^{\pi} \left(2 - \sin\left(\frac{\theta}{4}\right)\right)^2 \mathrm{d}\theta - \frac{1}{2}\int_{\pi}^{3\pi} \left(2 - \sin\left(\frac{\theta}{4}\right)\right)^2 \mathrm{d}\theta + \frac{1}{2}\int_{3\pi}^{4\pi} \left(2 - \sin\left(\frac{\theta}{4}\right)\right)^2 \mathrm{d}\theta$

$= 2\left(-\frac{17}{2} + 4\sqrt{2} + \frac{9\pi}{4}\right) - \left(1 - 8\sqrt{2} + \frac{9\pi}{2}\right)$

$= (-18 + 16\sqrt{2})$ units^2 as req.

22) a)

$z^n = (\cos\theta + \mathrm{i}\sin\theta)^n = \cos n\theta + \mathrm{i}\sin n\theta$

And so $z^n + z^{-n} = \cos n\theta + \mathrm{i}\sin n\theta + \cos(-n\theta) + \mathrm{i}\sin(-n\theta)$

$= \cos n\theta + \mathrm{i}\sin n\theta + \cos n\theta - \mathrm{i}\sin n\theta$

$= 2\cos n\theta$

22) b)

$\left(z + \frac{1}{z}\right)^4 = z^4 + 4z^3 \times \frac{1}{z} + 6z^2 \times \frac{1}{z^2} + 4z \times \frac{1}{z^3} + \frac{1}{z^4}$

$= z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4}$

$= \left(z^4 + \frac{1}{z^4}\right) + 4\left(z^2 + \frac{1}{z^2}\right) + 6$

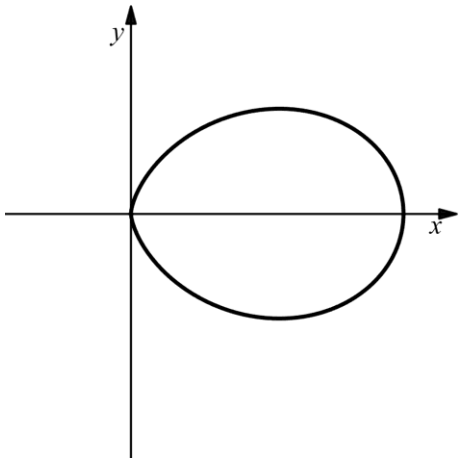
So

$(2\cos\theta)^4 = (2\cos 4\theta) + 4(2\cos 2\theta) + 6$

$16\cos^4\theta = 2\cos 4\theta + 8\cos 2\theta + 6$

$\cos^4\theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}$ as req.

22) c)



22) d)

$\frac{1}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(a\cos^2\theta)^2\,\mathrm{d}\theta = \frac{1}{2}a^2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^4\theta\,\mathrm{d}\theta$

Since $\cos^4\theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}$

$\frac{1}{2}a^2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^4\theta\,\mathrm{d}\theta = \frac{1}{2}a^2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}\right)\mathrm{d}\theta$

$= \left[\frac{1}{2}a^2\left(\frac{1}{32}\sin 4\theta + \frac{1}{4}\sin 2\theta + \frac{3}{8}\theta\right)\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$

$= \left(\frac{1}{2}a^2\left(0 + 0 + \frac{3}{8}\times\frac{\pi}{2}\right)\right) - \left(\frac{1}{2}a^2\left(0 + 0 + \frac{3}{8}\times -\frac{\pi}{2}\right)\right)$

$= \frac{3\pi}{16}a^2$ units^2

23) a)

$(x^2 + y^2)^3 = y^4$

$(r^2\cos^2\theta + r^2\sin^2\theta)^3 = r^4\sin^4\theta$

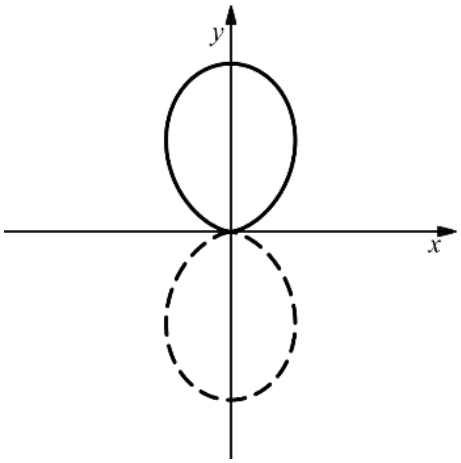
$r^6(\cos^2\theta + \sin^2\theta)^3 = r^4\sin^4\theta$

$r^6 = r^4\sin^4\theta$

$r^2 = \sin^4\theta$

23) b)

Since $r^2 = \sin^4\theta$ then consider $r = \pm\sin^2\theta$ and use dashed lines for negative values of r



23) c)

$\frac{1}{2}\int_0^{\pi}(\sin^2\theta)^2\,\mathrm{d}\theta = \frac{1}{2}\int_0^{\pi}\sin^4\theta\,\mathrm{d}\theta$

Since $\cos(2A) = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$

So $\sin^4 A = \left(\frac{1}{2} - \frac{1}{2}\cos(2A)\right)\left(\frac{1}{2} - \frac{1}{2}\cos(2A)\right) = \frac{1}{4} - \frac{1}{2}\cos(2A) + \frac{1}{4}\cos^2(2A)$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

So $\sin^4 A = \frac{1}{4} - \frac{1}{2}\cos(2A) + \frac{1}{4}\left(\frac{1}{2} + \frac{1}{2}\cos(4A)\right) = \frac{3}{8} - \frac{1}{2}\cos(2A) + \frac{1}{8}\cos(4A)$

$\frac{1}{2}\int_0^{\pi}\sin^4\theta\,\mathrm{d}\theta = \frac{1}{2}\int_0^{\pi}\left(\frac{3}{8} - \frac{1}{2}\cos(2\theta) + \frac{1}{8}\cos(4\theta)\right)\mathrm{d}\theta$

$= \left[\frac{1}{2}\left(\frac{3}{8}\theta - \frac{1}{4}\sin(2\theta) + \frac{1}{32}\sin(4\theta)\right)\right]_0^{\pi}$

$= \left(\frac{1}{2}\left(\frac{3}{8}\pi - 0 + 0\right)\right) - \left(\frac{1}{2}(0 - 0 + 0)\right)$

$= \frac{3}{16}\pi$

Two loops: $2 \times \frac{3}{16}\pi = \frac{3\pi}{8}$ units^2

24) a)

$(x^2 + y^2)^2 = a^2(x^2 - y^2)$

$(r^2\cos^2\theta + r^2\sin^2\theta)^2 = a^2(r^2\cos^2\theta - r^2\sin^2\theta)$

$r^4(\cos^2\theta + \sin^2\theta)^2 = a^2r^2(\cos^2\theta - \sin^2\theta)$

$r^4 = a^2r^2\cos 2\theta$

$r^2 = a^2\cos 2\theta$

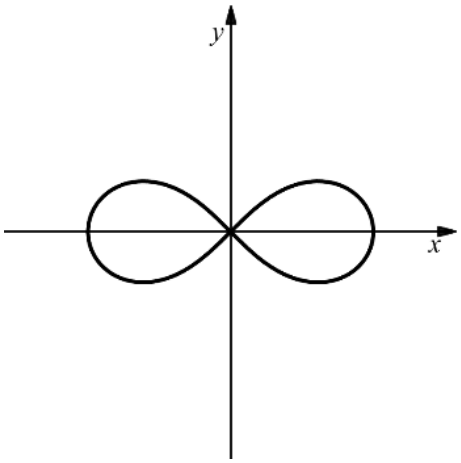
24) b)

Since $r^2 = a^2 \cos 2\theta$ then consider $r = \pm a\sqrt{\cos 2\theta}$ and use dashed lines for negative values of r

$y = \cos 2x$ is positive for $0 < x < \frac{\pi}{4}, \frac{3\pi}{4} < x < \frac{5\pi}{4}, \frac{7\pi}{4} < x < 2\pi$

$y = \cos 2x$ is negative for $\frac{\pi}{4} < x < \frac{3\pi}{4}, \frac{5\pi}{4} < x < \frac{7\pi}{4}$

So it only exists for $0 < x < \frac{\pi}{4}, \frac{3\pi}{4} < x < \frac{5\pi}{4}, \frac{7\pi}{4} < x < 2\pi$



24) c)

$\frac{1}{2}\int_{\frac{3\pi}{4}}^{\frac{5\pi}{4}} a^2 \cos 2\theta \, \mathrm{d}\theta = \frac{1}{2}a^2 \int_{\frac{3\pi}{4}}^{\frac{5\pi}{4}} \cos 2\theta \, \mathrm{d}\theta$

$= \left[\frac{1}{2}a^2\left(\frac{1}{2}\sin 2\theta\right)\right]_{\frac{3\pi}{4}}^{\frac{5\pi}{4}}$

$= \left(\frac{1}{4}a^2\left(\sin\frac{5\pi}{2}\right)\right) - \left(\frac{1}{4}a^2\left(\sin\frac{3\pi}{2}\right)\right)$

$= \left(\frac{1}{4}a^2\right) - \left(\frac{1}{4}a^2(-1)\right)$

$= \frac{1}{2}a^2$

Two loops: $2 \times \frac{1}{2}a^2 = a^2$ units^2

25) a)

$y^2 = a^2(x-y)^{\frac{4}{3}} - x^2$

$r^2\sin^2\theta = a^2(r\cos\theta - r\sin\theta)^{\frac{4}{3}} - r^2\cos^2\theta$

$(r^2\cos^2\theta + r^2\sin^2\theta)^3 = a^6(r\cos\theta - r\sin\theta)^4$

$r^6(\cos^2\theta + \sin^2\theta)^3 = a^6(r^2\cos^2\theta - 2r^2\sin\theta\cos\theta + r^2\sin^2\theta)^2$

$r^6 = a^6(r^2 - r^2\sin 2\theta)^2$

$r^6 = a^6r^4(1-\sin 2\theta)^2$

$r^2 = a^6(1-\sin 2\theta)^2$

25) b)

If $y = -x$ is a line of symmetry, then $(-y,-x) \mapsto (x,y)$

$(-x)^2 = a^2\big((-y)-(-x)\big)^{\frac{4}{3}} - (-y)^2$

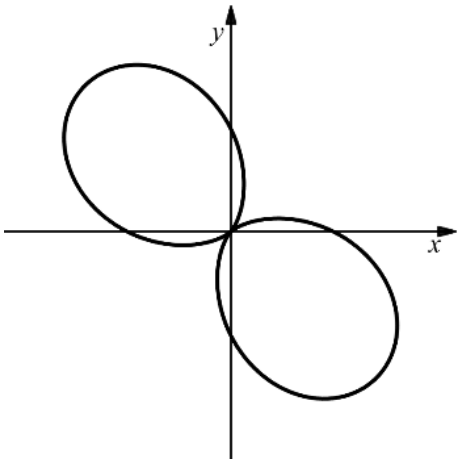
$x^2 = a^2(-y+x)^{\frac{4}{3}} - y^2$

which is $y^2 = a^2(x-y)^{\frac{4}{3}} - x^2$ so $y = -x$ is a line of symmetry

25) c)

Since $r^2 = a^6(1-\sin 2\theta)^2$ then consider $r = \pm a^3(1-\sin 2\theta)$ and use dashed lines for negative values of r

$y = 1 - \sin 2x$ is always positive, and $y = 0$ when $x = \frac{\pi}{4}$, $x = \frac{5\pi}{4}$



25) d)

$\frac{1}{2}\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} a^6(1-\sin 2\theta)^2\,\mathrm{d}\theta = \frac{1}{2}a^6\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}}(1-2\sin 2\theta+\sin^2(2\theta))\,\mathrm{d}\theta$

Since $\cos(2A) = 1-2\sin^2 A$ then $\sin^2 A = \frac{1}{2}-\frac{1}{2}\cos(2A)$

$\frac{1}{2}a^6\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}}(1-2\sin 2\theta+\sin^2(2\theta))\,\mathrm{d}\theta = \frac{1}{2}a^6\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}}\left(1-2\sin 2\theta+\left(\frac{1}{2}-\frac{1}{2}\cos(4\theta)\right)\right)\mathrm{d}\theta$

$= \frac{1}{2}a^6\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}}\left(\frac{3}{2}-2\sin 2\theta-\frac{1}{2}\cos(4\theta)\right)\mathrm{d}\theta$

$= \left[\frac{1}{2}a^6\left(\frac{3}{2}\theta-\cos 2\theta-\frac{1}{8}\sin(4\theta)\right)\right]_{\frac{\pi}{4}}^{\frac{5\pi}{4}}$

$= \left(\frac{1}{2}a^6\left(\frac{3}{2}\times\frac{5\pi}{4}-0-0\right)\right)-\left(\frac{1}{2}a^6\left(\frac{3}{2}\times\frac{\pi}{4}-0-0\right)\right)$

$= \frac{1}{2}a^6\times\frac{3}{2}\pi$

Two loops: $2\times\frac{3}{4}\pi a^6 = \frac{3}{2}\pi a^6$ units^2

26) a)

$(x^2+y^2)^2+ax(3y^2-x^2)=0$

$(r^2\cos^2\theta+r^2\sin^2\theta)^2+ar\cos\theta\,(3r^2\sin^2\theta-r^2\cos^2\theta)=0$

$r^4(\cos^2\theta+\sin^2\theta)^2+ar^3\cos\theta\,(3\sin^2\theta-\cos^2\theta)=0$

$r^4 = ar^3\cos\theta\,(\cos^2\theta-3\sin^2\theta)$

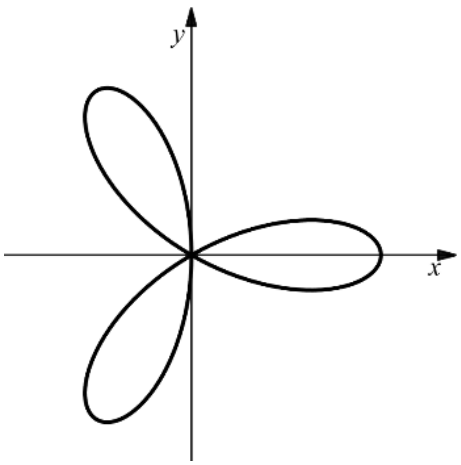
$r = a\cos\theta\left(\cos^2\theta-3(1-\cos^2\theta)\right)$

$r = a\cos\theta\,(4\cos^2\theta-3)$

$r = a(4\cos^3\theta-3\cos\theta)$

$r = a\cos 3\theta$ as req.

26) b)



26) c)

$\frac{1}{2}\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}}(a\cos 3\theta)^2\,\mathrm{d}\theta = \frac{1}{2}a^2\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}}\cos^2(3\theta)\,\mathrm{d}\theta$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

$\frac{1}{2}a^2\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}}\cos^2(3\theta)\,\mathrm{d}\theta = \frac{1}{2}a^2\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}}\left(\frac{1}{2} + \frac{1}{2}\cos(6\theta)\right)\mathrm{d}\theta$

$= \left[\frac{1}{2}a^2\left(\frac{1}{2}\theta + \frac{1}{12}\sin(6\theta)\right)\right]_{-\frac{\pi}{6}}^{\frac{\pi}{6}}$

$= \left(\frac{1}{2}a^2\left(\frac{\pi}{12} + 0\right)\right) - \left(\frac{1}{2}a^2\left(-\frac{\pi}{12} + 0\right)\right)$

$= \frac{\pi}{12}a^2$

Three loops: $3 \times \frac{\pi}{12}a^2 = \frac{\pi}{4}a^2$ units^2

27) a)

$\frac{1}{2}\int_0^{\pi}(3\cos\theta + \sin\theta)^2\mathrm{d}\theta = \frac{1}{2}\int_0^{\pi}(9\cos^2\theta + 6\sin\theta\cos\theta + \sin^2\theta)\mathrm{d}\theta$

$= \frac{1}{2}\int_0^{\pi}\left(9\cos^2\theta + 3\sin 2\theta + (1 - \cos^2\theta)\right)\mathrm{d}\theta$

$= \frac{1}{2}\int_0^{\pi}(8\cos^2\theta + 3\sin 2\theta + 1)\mathrm{d}\theta$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

$\frac{1}{2}\int_0^{\pi}(8\cos^2\theta + 3\sin 2\theta + 1)\mathrm{d}\theta = \frac{1}{2}\int_0^{\pi}\left(8\left(\frac{1}{2} + \frac{1}{2}\cos(2\theta)\right) + 3\sin 2\theta + 1\right)\mathrm{d}\theta$

$= \frac{1}{2}\int_0^{\pi}(5 + 4\cos 2\theta + 3\sin 2\theta)\mathrm{d}\theta$

$= \left[\frac{5}{2}\theta + \sin 2\theta - \frac{3}{4}\cos 2\theta\right]_0^{\pi}$

$= \left(\frac{5}{2}\pi + 0 - \frac{3}{4}\right) - \left(0 + 0 - \frac{3}{4}\right)$

$= \frac{5\pi}{2}$ units^2

27) b)

$\frac{\mathrm{d}r}{\mathrm{d}\theta} = -3\sin\theta + \cos\theta$ Max/min points when $-3\sin\theta + \cos\theta = 0$

$\therefore \tan\theta = \frac{1}{3}$ so $\sin\theta = \frac{1}{\sqrt{10}}$ and $\cos\theta = \frac{3}{\sqrt{10}}$

So $r = 3\cos\theta + \sin\theta = \frac{9}{\sqrt{10}} + \frac{1}{\sqrt{10}} = \frac{10}{\sqrt{10}}$

So $r\cos\theta = \frac{10}{\sqrt{10}} \times \frac{3}{\sqrt{10}} = 3$ and $r\sin\theta = \frac{10}{\sqrt{10}} \times \frac{1}{\sqrt{10}} = 1$

$\therefore (3, 1)$

27) c)

$r^2 = 3r\cos\theta + r\sin\theta$

$x^2 + y^2 = 3x + y$

$x^2 - 3x + y^2 - y = 0$

$\left(x - \frac{3}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 - \frac{9}{4} - \frac{1}{4} = 0$

$\left(x - \frac{3}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \frac{5}{2}$ which is a circle with centre $\left(\frac{3}{2}, -\frac{1}{2}\right)$ and radius $\sqrt{\frac{5}{2}}$

28)

$\frac{1}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(2(1-\sin\theta)\cos\theta)^2\,\mathrm{d}\theta = 2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(1-\sin\theta)^2\cos^2\theta\,\mathrm{d}\theta$

$= 2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(1-2\sin\theta+\sin^2\theta)\cos^2\theta\,\mathrm{d}\theta$

$= 2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^2\theta\,\mathrm{d}\theta - 4\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin\theta\cos^2\theta\,\mathrm{d}\theta + 2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin^2\theta\cos^2\theta\,\mathrm{d}\theta$

Consider each of the separate integrals:

$I_1 = 2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^2\theta\,\mathrm{d}\theta$

$I_2 = -4\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin\theta\cos^2\theta\,\mathrm{d}\theta$

$I_3 = 2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin^2\theta\cos^2\theta\,\mathrm{d}\theta$

Find I_1:

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

$2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^2\theta\,\mathrm{d}\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(1+\cos(2\theta))\,\mathrm{d}\theta = \left[\theta + \frac{1}{2}\sin(2\theta)\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$

$= \left(\frac{\pi}{2}+0\right) - \left(-\frac{\pi}{2}+0\right) = \pi$

Find I_2:

$-4\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin\theta\cos^2\theta\,\mathrm{d}\theta = -4\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin\theta\,(\cos\theta)^2\,\mathrm{d}\theta = \left[\frac{4}{3}\cos^3\theta\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$

$= (0) - (0) = 0$

Find I_3:

Since $2\sin^2\theta\cos^2\theta = \frac{1}{2}\times 4\sin^2\theta\cos^2\theta = \frac{1}{2}\times(\sin 2\theta)^2 = \frac{1}{2}\sin^2 2\theta$

$2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin^2\theta\cos^2\theta\,\mathrm{d}\theta = \frac{1}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin^2(2\theta)\,\mathrm{d}\theta$

Since $\cos(2A) = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$

$\frac{1}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin^2(2\theta)\,\mathrm{d}\theta = \frac{1}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{1}{2} - \frac{1}{2}\cos(4\theta)\right)\mathrm{d}\theta = \left[\frac{1}{4}\theta - \frac{1}{16}\sin(4\theta)\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$

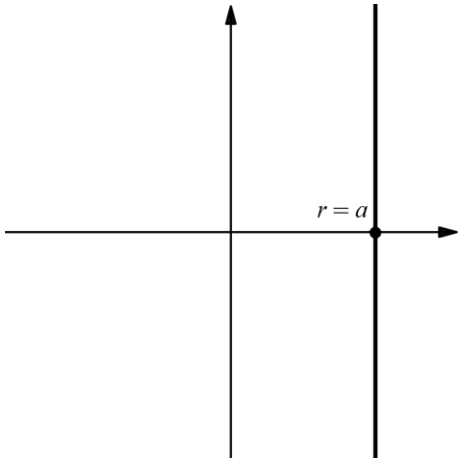
$= \left(\frac{1}{4}\times\frac{\pi}{2} - 0\right) - \left(\frac{1}{4}\times -\frac{\pi}{2} - 0\right) = \frac{\pi}{4}$

So the total area is:

$\pi + 0 + \frac{\pi}{4} = \frac{5\pi}{4}$ units^2 as req.

29) a)

$r = \frac{a}{\cos\theta} \Rightarrow r\cos\theta = a$ So $x = a$



29) b)

$\frac{1}{2}\int_0^k (a\sec\theta)^2 \mathrm{d}\theta = \frac{1}{2}a^2\int_0^k \sec^2\theta\,\mathrm{d}\theta$

$= \left[\frac{1}{2}a^2\tan\theta\right]_0^k$

$= \left(\frac{1}{2}a^2\tan k\right) - (0)$

$= \frac{1}{2}a^2\tan k$ units^2

30)

$\frac{1}{2}\int_0^{\frac{\pi}{4}} \left(a\sec^2\left(\frac{\theta}{2}\right)\right)^2 \mathrm{d}\theta = \frac{1}{2}a^2\int_0^{\frac{\pi}{4}} \sec^4\left(\frac{\theta}{2}\right)\mathrm{d}\theta$

But $\sec^4 x = \sec^2 x\sec^2 x = \sec^2 x\,(1+\tan^2 x)$

Let $u = \tan\frac{\theta}{2}$ so that $\frac{\mathrm{d}u}{\mathrm{d}\theta} = \frac{1}{2}\sec^2\left(\frac{\theta}{2}\right)$

When $\theta = 0$, $u = 0$, when $\theta = \frac{\pi}{4}$, $u = \tan\frac{\pi}{8}$

So

$\frac{1}{2}a^2\int_0^{\frac{\pi}{4}} \left(\sec^2\left(\frac{\theta}{2}\right)\left(1+\tan^2\left(\frac{\theta}{2}\right)\right)\right)\mathrm{d}\theta = a^2\int_0^{\tan\frac{\pi}{8}} (1+u^2)\mathrm{d}u$

$= \left[a^2\left(u + \frac{1}{3}u^3\right)\right]_0^{\tan\frac{\pi}{8}}$

$= \left(a^2\left(\tan\frac{\pi}{8} + \frac{1}{3}\tan^3\left(\frac{\pi}{8}\right)\right)\right) - (0)$

$= a^2\left(\tan\frac{\pi}{8} + \frac{1}{3}\tan^3\left(\frac{\pi}{8}\right)\right)$ units^{2}as req.

31) a)

$\frac{1}{2}\int_{-\pi}^{\pi}\left(2\sqrt{\cos\theta}\,e^{\frac{1}{3}\sin\theta}\right)^2 d\theta = 2\int_{-\pi}^{\pi}\cos\theta\,e^{\frac{2}{3}\sin\theta}d\theta$

Let $u = \sin\theta$ so $\frac{du}{d\theta} = \cos\theta$

When $\theta = -\frac{\pi}{2}$, $u = -1$, when $\theta = \frac{\pi}{2}$, $u = 1$

$2\int_{-\pi}^{\pi}\cos\theta\,e^{\frac{2}{3}\sin\theta}d\theta = 2\int_{-1}^{1}e^{\frac{2}{3}u}du = \left[3e^{\frac{2}{3}u}\right]_{-1}^{1}$

$= \left(3e^{\frac{2}{3}} - 3e^{-\frac{2}{3}}\right)$ units^2

31) b)

$r = 2(\cos\theta)^{\frac{1}{2}}e^{\frac{1}{3}\sin\theta}$

$\frac{dr}{d\theta} = \frac{2}{3}(\cos\theta)^{\frac{3}{2}}e^{\frac{1}{3}\sin\theta} - \sin\theta\,(\cos\theta)^{-\frac{1}{2}}e^{\frac{1}{3}\sin\theta}$

$= e^{\frac{1}{3}\sin\theta}\left(\frac{2}{3}(\cos\theta)^{\frac{3}{2}} - \sin\theta\,(\cos\theta)^{-\frac{1}{2}}\right)$

$= e^{\frac{1}{3}\sin\theta}\left(\frac{2\cos\theta\sqrt{\cos\theta}}{3} - \frac{\sin\theta}{\sqrt{\cos\theta}}\right)$

$= e^{\frac{1}{3}\sin\theta}\left(\frac{2\cos^2\theta}{3\sqrt{\cos\theta}} - \frac{3\sin\theta}{3\sqrt{\cos\theta}}\right)$

$= e^{\frac{1}{3}\sin\theta}\left(\frac{2\cos^2\theta - 3\sin\theta}{3\sqrt{\cos\theta}}\right)$

Since greatest value of r occurs when $\frac{dr}{d\theta} = 0$

$e^{\frac{1}{3}\sin\theta}\left(\frac{2\cos^2\theta - 3\sin\theta}{3\sqrt{\cos\theta}}\right) = 0$

$2\cos^2\theta - 3\sin\theta = 0$

$\Rightarrow 2(1-\sin^2\theta) - 3\sin\theta = 0 \Rightarrow 2\sin^2\theta + 3\sin\theta - 2 = 0$

$\Rightarrow (\sin\theta - 2)(2\sin\theta - 1) = 0$

$\therefore \sin\theta = \frac{1}{2}$ so that $\theta = \frac{\pi}{6} \Rightarrow \cos\theta = \frac{\sqrt{3}}{2}$

So $r = 2\left(\frac{\sqrt{3}}{2}\right)^{\frac{1}{2}}e^{\frac{1}{3}\left(\frac{1}{2}\right)} = 2\sqrt{\frac{\sqrt{3}}{2}}\,e^{\frac{1}{6}}$ as req.

32) a)

$\frac{1}{2}\int_0^{2\pi}\left(\frac{12}{k}+k(2+\sin\theta)\right)^2 \mathrm{d}\theta = \frac{1}{2}\int_0^{2\pi}\left(\frac{144}{k^2}+24(2+\sin\theta)+k^2(2+\sin\theta)^2\right)\mathrm{d}\theta$

$= \frac{1}{2}\int_0^{2\pi}\left(\frac{144}{k^2}+48+24\sin\theta+4k^2+4k^2\sin\theta+k^2\sin^2\theta\right)\mathrm{d}\theta$

Since $\cos(2A) = 1-2\sin^2 A$ then $\sin^2 A = \frac{1}{2}-\frac{1}{2}\cos(2A)$

$\frac{1}{2}\int_0^{2\pi}\left(\frac{144}{k^2}+48+24\sin\theta+4k^2+4k^2\sin\theta+k^2\left(\frac{1}{2}-\frac{1}{2}\cos(2\theta)\right)\right)\mathrm{d}\theta$

$= \int_0^{2\pi}\left(\frac{72}{k^2}+24+\frac{9}{4}k^2+(2k^2+12)\sin\theta-\frac{1}{4}k^2\cos(2\theta)\right)\mathrm{d}\theta$

$= \left[\left(\frac{72}{k^2}+24+\frac{9}{4}k^2\right)\theta-(2k^2+12)\cos\theta-\frac{1}{8}k^2\sin(2\theta)\right]_0^{2\pi}$

$= \left(\left(\frac{72}{k^2}+24+\frac{9}{4}k^2\right)2\pi-(2k^2+12)-0\right)-(0-(2k^2+12)-0)$

$= \left(\frac{72}{k^2}+24+\frac{9}{4}k^2\right)2\pi$

$= \left(\frac{144}{k^2}+48+\frac{9}{2}k^2\right)2\pi$ units^2 as req.

32) b)

$A = \left(\frac{144}{k^2}+48+\frac{9}{2}k^2\right)2\pi = \left(144k^{-2}+48+\frac{9}{2}k^2\right)2\pi$

$\frac{\mathrm{d}A}{\mathrm{d}k} = (-288k^{-3}+9k)2\pi$

Minimum occurs when $\frac{\mathrm{d}A}{\mathrm{d}k} = 0$

$(-288k^{-3}+9k)2\pi = 0$ so $9k-\frac{288}{k^3} = 0$

$9k = \frac{288}{k^3}$

$k^4 = 32$

$k = \pm 32^{\frac{1}{4}} = \pm\sqrt[4]{32}$

But k is a positive constant, so $k = \sqrt[4]{32}$

33) a)

$\theta = 0$ and $\theta = \frac{\pi}{2}$ so $r = \cos 0$ and $r = \cos\frac{\pi}{2}$

$\therefore (1,0)$ and $\left(0,\frac{\pi}{2}\right)$

33) b)

Consider $r(\theta) = r(-\theta)$

C_1: $\cos(-\theta) = \cos\theta = r$

C_2: $a(1 + \cos(-\theta)) = a(1 + \cos\theta) = r$

So both curves are symmetrical about the initial line.

33) c)

$\frac{1}{2}\int \left(a(1 + \cos\theta)\right)^2 \mathrm{d}\theta = \frac{1}{2}a^2 \int (1 + 2\cos\theta + \cos^2\theta)\mathrm{d}\theta$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

$\frac{1}{2}a^2 \int (1 + 2\cos\theta + \cos^2\theta)\mathrm{d}\theta = \frac{1}{2}a^2 \int \left(1 + 2\cos\theta + \frac{1}{2} + \frac{1}{2}\cos(2\theta)\right)\mathrm{d}\theta$

$= \frac{1}{2}a^2 \int \left(\frac{3}{2} + 2\cos\theta + \frac{1}{2}\cos(2\theta)\right)\mathrm{d}\theta$

$= \frac{1}{2}a^2 \left(\frac{3}{2}\theta + 2\sin\theta + \frac{1}{4}\sin(2\theta)\right)(+c)$

So that

$\frac{1}{2}\int_0^{2\pi} \left(a(1 + \cos\theta)\right)^2 \mathrm{d}\theta = \left[\frac{1}{2}a^2 \left(\frac{3}{2}\theta + 2\sin\theta + \frac{1}{4}\sin(2\theta)\right)\right]_0^{2\pi}$

$= \left(\frac{1}{2}a^2 \left(\frac{3}{2} \times 2\pi + 0 + 0\right)\right) - \left(\frac{1}{2}a^2(0 + 0 + 0)\right)$

$= \frac{3}{2}a^2\pi$

Area of C_1 is $\pi r^2 = \pi\left(\frac{1}{2}\right)^2 = \frac{1}{4}\pi$

So the region R is:

$\left(\frac{3}{2}a^2\pi - \frac{1}{4}\pi\right)$ units^2

34) a)

$r = 2.1 + a\cos 2\theta$ has a max of $r = 3$ when $\cos 2\theta = 1$ and so $a = 0.9$

34) b)

$\frac{1}{2}\int_0^{2\pi}(2.1+0.9\cos 2\theta)^2\mathrm{d}\theta = \frac{1}{2}\int_{-\pi}^{\pi}(4.41+3.78\cos 2\theta+0.81\cos^2 2\theta)\mathrm{d}\theta$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2}+\frac{1}{2}\cos(2A)$

$\frac{1}{2}\int_{-\pi}^{\pi}\left(4.41+3.78\cos 2\theta+0.81\left(\frac{1}{2}+\frac{1}{2}\cos(4\theta)\right)\right)\mathrm{d}\theta$

$= \frac{1}{2}\int_{-\pi}^{\pi}(2.4075+1.89\cos 2\theta+0.2025\cos(4\theta))\mathrm{d}\theta$

$= [(2.4075\theta+0.945\sin 2\theta+0.050625\sin(4\theta))]_0^{2\pi}$

$= \big((2.4075\times 2\pi+0+0)\big)-\big((0+0+0)\big)$

$= 4.815\pi$

Area of circle: $\pi\times 3^2 = 9\pi$

So shaded region is $\frac{837}{200}\pi$ units^2

34) c)

$\frac{2\times 2^2\times 4.185\pi}{40} = 2.62951\ldots \therefore$ 3 tins of paint.

35) a)

P is when $\theta = \frac{\pi}{2}$ and $r = 1.3$

So $1.3 = 1+a\sin\left(3\times\frac{\pi}{2}\right) \Rightarrow 0.3 = -a \therefore a = -0.3$

35) b)

$\frac{1}{2}\int_0^{2\pi}(1-0.3\sin(3\theta))^2\mathrm{d}\theta = \frac{1}{2}\int_0^{2\pi}(1-0.6\sin(3\theta)+0.09\sin^2(3\theta))\mathrm{d}\theta$

Since $\cos(2A) = 1-2\sin^2 A$ then $\sin^2 A = \frac{1}{2}-\frac{1}{2}\cos(2A)$

$\int_0^{2\pi}(0.5-0.3\sin(3\theta)+0.045\sin^2(3\theta))\mathrm{d}\theta$

$= \int_0^{2\pi}\left(0.5-0.3\sin(3\theta)+0.045\left(\frac{1}{2}-\frac{1}{2}\cos(6\theta)\right)\right)\mathrm{d}\theta$

$= \int_0^{2\pi}(0.5225-0.3\sin(3\theta)-0.0225\cos(6\theta))\mathrm{d}\theta$

$= [0.5225\theta+0.1\cos(\theta)-0.00375\sin(6\theta)]_0^{2\pi}$

$= (0.5225\times 2\pi+0.1-0)-(0+0.1-0)$

$= 1.045\pi$

Area of square is $2.6^2 = 6.76$

So area left over is $6.76-1.045\pi = 3.48$ m^2

36) a)

P occurs when $\theta = \frac{\pi}{6}$ so $\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$

The x-coordinate of P is $\left(a + b\cos\frac{\pi}{6}\right)\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}\left(a + \frac{\sqrt{3}}{2}b\right)$

The line going through P and Q is a straight line which is of the form $r\cos\theta = k$ for some positive constant k

$r\cos\theta = \frac{\sqrt{3}}{2}\left(a + \frac{\sqrt{3}}{2}b\right)$

$\therefore r = \frac{\sqrt{3}}{2}\left(a + \frac{\sqrt{3}}{2}b\right)\sec\theta$

36) b)

$2 = a + \cos\frac{\pi}{6}$ so $a = 2 - \frac{\sqrt{3}}{2}$

36) c)

Area of triangle:

$\frac{1}{2} \times 2 \times 2 \times \sin\frac{\pi}{3} = \sqrt{3}$

Area of $r = a + b\cos\theta$:

$\frac{1}{2}\int_{-\pi}^{\pi}\left(2 - \frac{\sqrt{3}}{2} + \cos\theta\right)^2 \mathrm{d}\theta = \frac{1}{2}\int_{-\pi}^{\pi}\left(\left(2 - \frac{\sqrt{3}}{2}\right)^2 + \left(4 - \sqrt{3}\right)\cos\theta + \cos^2\theta\right)\mathrm{d}\theta$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

$\frac{1}{2}\int_{-\pi}^{\pi}\left(\left(2 - \frac{\sqrt{3}}{2}\right)^2 + \left(4 - \sqrt{3}\right)\cos\theta + \cos^2\theta\right)\mathrm{d}\theta$

$= \frac{1}{2}\int_{-\pi}^{\pi}\left(\left(2 - \frac{\sqrt{3}}{2}\right)^2 + \left(4 - \sqrt{3}\right)\cos\theta + \left(\frac{1}{2} + \frac{1}{2}\cos(2\theta)\right)\right)\mathrm{d}\theta$

$= \int_{-\pi}^{\pi}\left(\frac{1}{2}\left(2 - \frac{\sqrt{3}}{2}\right)^2 + \frac{1}{4} + \left(4 - \sqrt{3}\right)\cos\theta + \frac{1}{4}\cos(2\theta)\right)\mathrm{d}\theta$

$= \left[\left(\frac{1}{8}\left(21 - 8\sqrt{3}\right)\right)\theta + \left(4 - \sqrt{3}\right)\sin\theta + \frac{1}{8}\sin(2\theta)\right]_{-\pi}^{\pi}$

$= \left(\left(\frac{1}{8}\left(21 - 8\sqrt{3}\right)\right)\pi + 0 + 0\right) - \left(\left(\frac{1}{8}\left(21 - 8\sqrt{3}\right)\right)(-\pi) + 0 + 0\right)$

$= \frac{1}{4}\left(21 - 8\sqrt{3}\right)\pi$

So the percentage is

$\frac{\sqrt{3}}{\frac{1}{4}\left(21 - 8\sqrt{3}\right)\pi} \times 100 = 30.9\%$ (3 SF)

37) a)

When $\theta = \frac{\pi}{4}$ then $r = 1 + \tan\frac{\pi}{4} = 2$ so OQ has length 2 units

37) b)

Area of triangle: $\frac{1}{2} \times 2 \times 3 \times \sin\frac{\pi}{4} = \frac{3\sqrt{2}}{2}$

Polar area:

$\frac{1}{2}\int_0^{\frac{\pi}{4}}(1 + \tan\theta)^2 \mathrm{d}\theta = \frac{1}{2}\int_0^{\frac{\pi}{4}}(1 + 2\tan\theta + \tan^2\theta)\mathrm{d}\theta$

$= \frac{1}{2}\int_0^{\frac{\pi}{4}}(2\tan\theta + \sec^2\theta)\mathrm{d}\theta$

$= \int_0^{\frac{\pi}{4}}\left(\tan\theta + \frac{1}{2}\sec^2\theta\right)\mathrm{d}\theta$

$= \left[\ln|\sec\theta| + \frac{1}{2}\tan\theta\right]_0^{\frac{\pi}{4}}$

$= \left(\ln\left|\sec\frac{\pi}{4}\right| + \frac{1}{2}\tan\frac{\pi}{4}\right) - \left(\ln|\sec 0| + \frac{1}{2}\tan 0\right)$

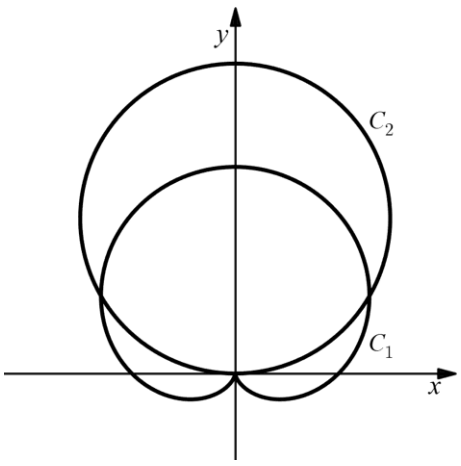
$= \left(\ln\sqrt{2} + \frac{1}{2}\right) - (0 + 0)$

$= \frac{1}{2}\ln 2 + \frac{1}{2}$

So the shaded area is:

$\frac{3\sqrt{2}}{2} - \left(\frac{1}{2}\ln 2 + \frac{1}{2}\right) = \frac{1}{2}\left(3\sqrt{2} - 1 - \ln 2\right)$ units^2 as req.

38) a)



38) b)

$1 + \sin\theta = 3\sin\theta$

$\sin\theta = \frac{1}{2}$ so $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$

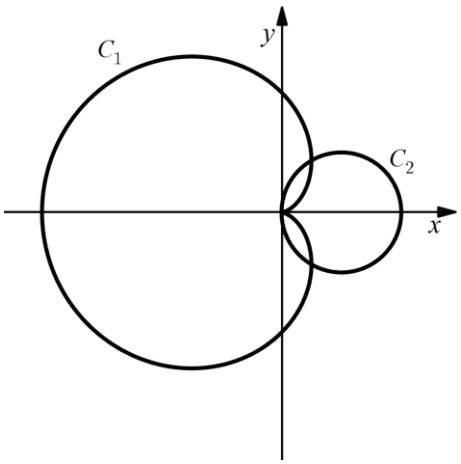
So $r = 1 + \sin\frac{\pi}{6} = \frac{3}{2}$ and $r = 1 + \sin\frac{5\pi}{6} = \frac{3}{2}$

So $\left(\frac{3}{2}, \frac{\pi}{6}\right)$ and $\left(\frac{3}{2}, \frac{5\pi}{6}\right)$

38) c)

Both curves go through the pole, but C_1 does when $\theta = \frac{3\pi}{2}$ and C_2 does when $\theta = 0$ so they do not intersect at the pole.

39) a)



39) b)

$2(1 - \cos\theta) = 2\cos\theta$

$2 - 2\cos\theta = 2\cos\theta$

$\cos\theta = \frac{1}{2}$ so $\theta = \frac{\pi}{3}$ or $\theta = -\frac{\pi}{3}$

So $r = 2\cos\frac{\pi}{3} = 1$ and $r = 2\cos\left(-\frac{\pi}{3}\right) = 1$

So $\left(1, \frac{\pi}{3}\right)$ and $\left(1, -\frac{\pi}{3}\right)$

39) c)

Both curves go through the pole, but C_1 does when $\theta = 0$ and C_2 does when $\theta = -\frac{\pi}{2}$ so they do not intersect at the pole.

40)

$1-\cos\theta = 1+\cos\theta \Rightarrow \cos\theta = 0$ so $\theta = \frac{\pi}{2}$ or $\theta = \frac{3\pi}{2}$

Find the area:

$\frac{1}{2}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} r_1^2 \mathrm{d}\theta - \frac{1}{2}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} r_2^2 \mathrm{d}\theta = \frac{1}{2}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (r_1^2 - r_2^2)\mathrm{d}\theta$

So that

$\frac{1}{2}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} ((1-\cos\theta)^2 - (1+\cos\theta)^2)\mathrm{d}\theta$

$= \frac{1}{2}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (1 - 2\cos\theta + \cos^2\theta - 1 - 2\cos\theta - \cos^2\theta)\mathrm{d}\theta$

$= -\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} 2\cos\theta \,\mathrm{d}\theta$

$= [-2\sin\theta]_{\frac{\pi}{2}}^{\frac{3\pi}{2}}$

$= -2\sin\frac{3\pi}{2} + 2\sin\frac{\pi}{2}$

$= 4$ units^2

41)

$\cos\theta\sqrt{2\sin\theta} = 3^{-\frac{1}{4}}\sin\theta \Rightarrow 2\cos^2\theta\sin\theta = \frac{1}{\sqrt{3}}\sin^2\theta \Rightarrow \sin^2\theta - 2\sqrt{3}\cos^2\theta\sin\theta = 0$

$\Rightarrow \sin^2\theta - 2\sqrt{3}(1-\sin^2\theta)\sin\theta = 0 \Rightarrow \sin^2\theta - 2\sqrt{3}\sin\theta + 2\sqrt{3}\sin^3\theta = 0$

$\Rightarrow \sin\theta\left(2\sqrt{3}\sin^2\theta + \sin\theta - 2\sqrt{3}\right) = 0$

So $\sin\theta = 0$ or $\sin\theta = \frac{\sqrt{3}}{2}$ (note: $\sin\theta = -\frac{2\sqrt{3}}{3}$ is not possible)

So $\theta = 0$ or $\theta = \frac{\pi}{3}$

Find the area:

$\frac{1}{2}\int_0^{\frac{\pi}{3}}\left(3^{-\frac{1}{4}}\sin\theta\right)^2 \mathrm{d}\theta + \frac{1}{2}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\left(\cos\theta\sqrt{2\sin\theta}\right)^2\mathrm{d}\theta$

Since

$\frac{1}{2}\int \left(3^{-\frac{1}{4}}\sin\theta\right)^2 \mathrm{d}\theta = \frac{1}{2\sqrt{3}}\int \sin^2\theta \ \mathrm{d}\theta = \int \left(\frac{1}{4\sqrt{3}} - \frac{1}{4\sqrt{3}}\cos 2\theta\right) \mathrm{d}\theta$

$= \frac{1}{4\sqrt{3}}\theta - \frac{1}{8\sqrt{3}}\sin 2\theta \ (+c)$

And

$\frac{1}{2}\int \left(\cos\theta\sqrt{2\sin\theta}\right)^2\mathrm{d}\theta = \int (\cos^2\theta\sin\theta)\mathrm{d}\theta = -\frac{1}{3}\cos^3\theta \ (+c)$

Then

$\frac{1}{2}\int_0^{\frac{\pi}{3}}\left(3^{-\frac{1}{4}}\sin\theta\right)^2 \mathrm{d}\theta + \frac{1}{2}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\left(\cos\theta\sqrt{2\sin\theta}\right)^2\mathrm{d}\theta$

$= \left[\frac{1}{4\sqrt{3}}\theta - \frac{1}{8\sqrt{3}}\sin 2\theta\right]_0^{\frac{\pi}{3}} + \left[-\frac{1}{3}\cos^3\theta\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$

$= \left(\left(\frac{\pi}{12\sqrt{3}} - \frac{1}{16}\right) - (0-0)\right) + \left((0) - \left(-\frac{1}{24}\right)\right)$

$= \left(\frac{\pi}{12\sqrt{3}} - \frac{1}{48}\right)$ units^2

42)

$4\sin\theta = 3 - 2\sin\theta \Rightarrow 4\sin\theta = 3 - 2\sin\theta \Rightarrow \sin\theta = \frac{1}{2}$ so $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$

Find the area:

$\frac{1}{2}\int_0^{\frac{\pi}{6}}(4\sin\theta)^2\mathrm{d}\theta + \frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}}(3-2\sin\theta)^2\mathrm{d}\theta + \frac{1}{2}\int_{\frac{5\pi}{6}}^{\pi}(4\sin\theta)^2\mathrm{d}\theta$

Since

$\frac{1}{2}\int(4\sin\theta)^2\mathrm{d}\theta = \frac{1}{2}\int 16\left(\frac{1}{2} - \frac{1}{2}\cos(2\theta)\right)\mathrm{d}\theta = \int 4 - 4\cos(2\theta)\,\mathrm{d}\theta = 4\theta - 2\sin 2\theta\ (+c)$

And $\frac{1}{2}\int(3-2\sin\theta)^2\mathrm{d}\theta = \frac{1}{2}\int(9 - 12\sin\theta + 4\sin^2\theta)\mathrm{d}\theta$

$= \int\left(\frac{11}{2} - 6\sin\theta + \cos(2\theta)\right)\mathrm{d}\theta = \frac{11}{2}\theta + 6\cos\theta + \frac{1}{2}\sin(2\theta)\ (+c)$

Then

$\frac{1}{2}\int_0^{\frac{\pi}{6}}(4\sin\theta)^2\mathrm{d}\theta + \frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}}(3-2\sin\theta)^2\mathrm{d}\theta + \frac{1}{2}\int_{\frac{5\pi}{6}}^{\pi}(4\sin\theta)^2\mathrm{d}\theta$

$= [4\theta - 2\sin 2\theta]_0^{\frac{\pi}{6}} + \left[\frac{11}{2}\theta + 6\cos\theta + \frac{1}{2}\sin(2\theta)\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}} + [4\theta - 2\sin 2\theta]_{\frac{5\pi}{6}}^{\pi}$

$= \left(\left(\frac{2\pi}{3} - \sqrt{3}\right) - (0-0)\right) + \left(\left(\frac{55\pi}{12} - 3\sqrt{3} - \frac{\sqrt{3}}{4}\right) - \left(\frac{11\pi}{12} + 3\sqrt{3} + \frac{\sqrt{3}}{4}\right)\right) + \left((4\pi - 0) - \left(\frac{10\pi}{3} + \sqrt{3}\right)\right)$

$= \left(5\pi - \frac{15\sqrt{3}}{2}\right)$ units^2

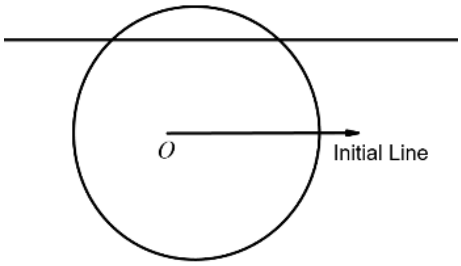
43) a)

When $\theta = \frac{2\pi}{3}$, $\frac{7\sqrt{3}}{4}\operatorname{cosec}\theta = \frac{7}{2}$ and $4 + \cos\theta = \frac{7}{2}$

When $\theta = \frac{2\pi}{3}$, $\frac{7\sqrt{3}}{4}\operatorname{cosec}\theta = 4.77243\ldots$ and $4 + \cos\theta = 4.77241\ldots$ (which are roughly the same)

So $\theta = \frac{2\pi}{3}$ and $\theta \approx 0.68817$ are points of intersection.

43) b)



43) c)

$\frac{1}{2}\int_0^{0.68817} r_1^2\,\mathrm{d}\theta + \frac{1}{2}\int_{0.68817}^{\frac{2\pi}{3}}(r_1^2 - r_2^2)\,\mathrm{d}\theta + \frac{1}{2}\int_{\frac{2\pi}{3}}^{\pi} r_1^2\,\mathrm{d}\theta$

$= \frac{1}{2}\int_0^{0.68817}(4+\cos\theta)^2\,\mathrm{d}\theta + \frac{1}{2}\int_{0.68817}^{\frac{2\pi}{3}}\left(\frac{7\sqrt{3}}{4}\operatorname{cosec}\theta\right)^2\mathrm{d}\theta + \frac{1}{2}\int_{\frac{2\pi}{3}}^{\pi}(4+\cos\theta)^2\,\mathrm{d}\theta$

Since

$\frac{1}{2}\int(4+\cos\theta)^2\mathrm{d}\theta = \frac{1}{2}\int(16+8\cos\theta+\cos^2\theta)\mathrm{d}\theta = \int\left(\frac{33}{4}+4\cos\theta+\frac{1}{4}\cos 2\theta\right)\mathrm{d}\theta$

$= \frac{33}{4}\theta + 4\sin\theta + \frac{1}{2}\sin 2\theta\ (+c)$

And

$\frac{1}{2}\int\left(\frac{7\sqrt{3}}{4}\operatorname{cosec}\theta\right)^2\mathrm{d}\theta = \int\frac{147}{32}\operatorname{cosec}^2\theta\,\mathrm{d}\theta = -\frac{147}{32}\cot\theta\ (+c)$

So that

$= \frac{1}{2}\int_0^{0.68817}(4+\cos\theta)^2\,\mathrm{d}\theta + \frac{1}{2}\int_{0.68817}^{\frac{2\pi}{3}}\left(\frac{7\sqrt{3}}{4}\operatorname{cosec}\theta\right)^2\mathrm{d}\theta + \frac{1}{2}\int_{\frac{2\pi}{3}}^{\pi}(4+\cos\theta)^2\,\mathrm{d}\theta$

$= \left[\frac{33}{4}\theta + 4\sin\theta + \frac{1}{2}\sin 2\theta\right]_0^{0.68817} + \left[-\frac{147}{32}\cot\theta\right]_{0.68817}^{\frac{2\pi}{3}} + \left[\frac{33}{4}\theta + 4\sin\theta + \frac{1}{2}\sin 2\theta\right]_{\frac{2\pi}{3}}^{\pi}$

$= \left(\frac{33}{4}\times 0.68817 + 4\sin 0.68817 + \frac{1}{2}\sin 1.37634\right) + \left(-\frac{147}{32}\cot\frac{2\pi}{3} + \frac{147}{32}\cot 0.68817\right) +$

$\left(\frac{33\pi}{4} + 0 + 0 - \frac{11\pi}{2} - 2\sqrt{3} - \frac{\sqrt{3}}{4}\right)$

$= 22$ units^2

44) a)

$3 + 2\sin\theta = 2 \Rightarrow 2\sin\theta = -1 \Rightarrow \sin\theta = -\frac{1}{2}$ so $\theta = -\frac{\pi}{6}$ or $\theta = \frac{7\pi}{6}$

$$\frac{1}{2}\int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}}(r_1^2 - r_2^2)\,\mathrm{d}\theta = \frac{1}{2}\int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}}((3+2\sin\theta)^2 - 2^2)\,\mathrm{d}\theta = \frac{1}{2}\int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}}(5 + 12\sin\theta + 4\sin^2\theta)\,\mathrm{d}\theta$$

$$= \frac{1}{2}\int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}}\left(5 + 12\sin\theta + 4\left(\frac{1}{2} - \frac{1}{2}\cos(2\theta)\right)\right)\mathrm{d}\theta = \int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}}\left(\frac{7}{2} + 6\sin\theta - \cos(2\theta)\right)\mathrm{d}\theta$$

$$= \left[\frac{7}{2}\theta - 6\cos\theta - \frac{1}{2}\sin(2\theta)\right]_{-\frac{\pi}{6}}^{\frac{7\pi}{6}}$$

$$= \left(\frac{49\pi}{12} + 3\sqrt{3} - \frac{\sqrt{3}}{4}\right) - \left(-\frac{7\pi}{12} - 3\sqrt{3} + \frac{\sqrt{3}}{4}\right)$$

$$= \left(\frac{11\sqrt{3}}{2} + \frac{14\pi}{3}\right)\text{ units}^2$$

44) b)

$4 - 2\cos\theta = 6\cos\theta \Rightarrow 8\cos\theta = 4 \Rightarrow \cos\theta = \frac{1}{2}$ so $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$

$$\frac{1}{2}\int_{\frac{\pi}{3}}^{\pi} r_1^2\,\mathrm{d}\theta - \frac{1}{2}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} r_2^2\,\mathrm{d}\theta$$

$$= \frac{1}{2}\int_{\frac{\pi}{3}}^{\pi}(4 - 2\cos\theta)^2\,\mathrm{d}\theta - \frac{1}{2}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}(6\cos\theta)^2\,\mathrm{d}\theta$$

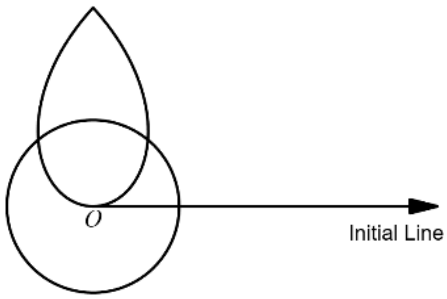
$$= \int_{\frac{\pi}{3}}^{\pi}(9 - 8\cos\theta + \cos 2\theta)\,\mathrm{d}\theta - \int_{\frac{\pi}{3}}^{\frac{\pi}{2}}(9 + 9\cos 2\theta)\,\mathrm{d}\theta$$

$$= \left[9\theta - 8\sin\theta + \frac{1}{2}\sin 2\theta\right]_{\frac{\pi}{3}}^{\pi} - \left[9\theta + \frac{9}{2}\sin 2\theta\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$= (9\pi - 0 + 0) - \left(3\pi - 4\sqrt{3} + \frac{\sqrt{3}}{4}\right) - \left(\left(\frac{9\pi}{2} + 0\right) - \left(3\pi + \frac{9\sqrt{3}}{4}\right)\right)$$

$$= \left(6\sqrt{3} + \frac{9\pi}{2}\right)\text{ units}^2$$

45) a)



45) b)

$2 = 2\sinh\theta \Rightarrow \sinh\theta = 1$ so $\theta = \sinh^{-1}(1) = \ln\left(1+\sqrt{2}\right)$ so only one point of intersection, but the region is symmetric across $\theta = \frac{\pi}{2}$

Find the half area:

$\frac{1}{2}\int_0^{\ln(1+\sqrt{2})}(2\sinh\theta)^2\mathrm{d}\theta + \frac{1}{2}\int_{\ln(1+\sqrt{2})}^{\frac{\pi}{2}}(2)^2\mathrm{d}\theta$

$= \frac{1}{2}\int_0^{\ln(1+\sqrt{2})}(4\sinh^2\theta)\,\mathrm{d}\theta + \frac{1}{2}\int_{\ln(1+\sqrt{2})}^{\frac{\pi}{2}}4\,\mathrm{d}\theta$

But since $\sinh^2\theta = \frac{1}{2}\cosh 2\theta - \frac{1}{2}$ then

$= \frac{1}{2}\int_0^{\ln(1+\sqrt{2})}(2\cosh 2\theta - 2)\,\mathrm{d}\theta + \frac{1}{2}\int_{\ln(1+\sqrt{2})}^{\frac{\pi}{2}}4\,\mathrm{d}\theta$

$= \int_0^{\ln(1+\sqrt{2})}(\cosh 2\theta - 1)\,\mathrm{d}\theta + \int_{\ln(1+\sqrt{2})}^{\frac{\pi}{2}}2\,\mathrm{d}\theta$

$= \left[\frac{1}{2}\sinh 2\theta - \theta\right]_0^{\ln(1+\sqrt{2})} + [2\theta]_{\ln(1+\sqrt{2})}^{\frac{\pi}{2}}$

$= \left(\left(\sqrt{2} - \ln\left(1+\sqrt{2}\right)\right) - (0-0)\right) + \left((\pi) - \left(2\ln\left(1+\sqrt{2}\right)\right)\right)$

$= \sqrt{2} + \pi - 3\ln\left(1+\sqrt{2}\right)$

So the total area is

$2\sqrt{2} + 2\pi - 6\ln\left(1+\sqrt{2}\right)$ units^2 as req.

46)

$2\sin\theta = \frac{2\sqrt{3}}{3}(1-\cos\theta) \Rightarrow \sqrt{3}\sin\theta = 1-\cos\theta \Rightarrow \sqrt{3}\sin\theta + \cos\theta = 1$

Since $R\cos(\theta-\alpha) = R\cos\theta\cos\alpha + R\sin\theta\sin\alpha$

Then $R = \sqrt{1+3} = 2$ and $\alpha = \arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$

So solve $2\cos\left(\theta - \frac{\pi}{3}\right) = 1 \Rightarrow \cos\left(\theta - \frac{\pi}{3}\right) = \frac{1}{2} \therefore \theta = 0$ or $\theta = \frac{2\pi}{3}$

Find the area:

$$\frac{1}{2}\int_0^{\frac{2\pi}{3}}\left(\frac{2\sqrt{3}}{3}(1-\cos\theta)\right)^2 \mathrm{d}\theta + \frac{1}{2}\int_{\frac{2\pi}{3}}^{\pi}(2\sin\theta)^2\mathrm{d}\theta$$

Since

$$\frac{1}{2}\int\left(\frac{2\sqrt{3}}{3}(1-\cos\theta)\right)^2 \mathrm{d}\theta = \frac{2}{3}\int(1-2\cos\theta+\cos^2\theta)\mathrm{d}\theta$$

$$= \int\left(1-\frac{4}{3}\cos\theta+\frac{1}{3}\cos(2\theta)\right)\mathrm{d}\theta = \theta - \frac{4}{3}\sin\theta + \frac{1}{6}\sin(2\theta)\ (+c)$$

And

$$\frac{1}{2}\int(2\sin\theta)^2\mathrm{d}\theta = \frac{1}{2}\int 4\left(\frac{1}{2}-\frac{1}{2}\cos(2\theta)\right)\mathrm{d}\theta = \int 1-\cos(2\theta)\,\mathrm{d}\theta = \theta - \frac{1}{2}\sin 2\theta\ (+c)$$

Then

$$\frac{1}{2}\int_0^{\frac{2\pi}{3}}\left(\frac{2\sqrt{3}}{3}(1-\cos\theta)\right)^2 \mathrm{d}\theta + \frac{1}{2}\int_{\frac{2\pi}{3}}^{\pi}(2\sin\theta)^2\mathrm{d}\theta$$

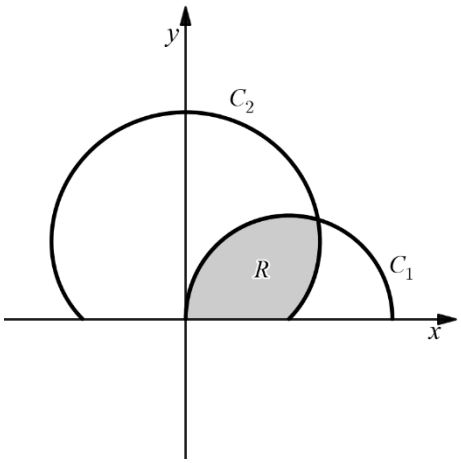
$$= \left[\theta - \frac{4}{3}\sin\theta + \frac{1}{6}\sin(2\theta)\right]_0^{\frac{2\pi}{3}} + \left[\theta - \frac{1}{2}\sin 2\theta\right]_{\frac{2\pi}{3}}^{\pi}$$

$$= \left(\left(\frac{2\pi}{3} - \frac{2\sqrt{3}}{3} - \frac{\sqrt{3}}{12}\right) - (0-0+0)\right) + \left((\pi-0) - \left(\frac{2\pi}{3}+\frac{\sqrt{3}}{4}\right)\right)$$

$$= \left(\pi - \sqrt{3}\right) \text{ units}^2$$

47)

First sketch to identify the region:



Point of intersection:

$2\cos\theta = 1 + \sin\theta \Rightarrow 2\cos\theta - \sin\theta = 1$

Since $R\cos(\theta + \alpha) = R\cos\theta\cos\alpha - R\sin\theta\sin\alpha$

Then $R = \sqrt{4+1} = \sqrt{5}$ and $\alpha = \arctan\left(\frac{1}{2}\right)$

So solve $\sqrt{5}\cos\left(\theta + \arctan\left(\frac{1}{2}\right)\right) = 1 \Rightarrow \cos\left(\theta + \arctan\left(\frac{1}{2}\right)\right) = \frac{1}{\sqrt{5}}$

We want an acute angle for the point of intersection so:

$\phi = \arccos\left(\frac{1}{\sqrt{5}}\right) - \arctan\left(\frac{1}{2}\right)$

Let $A = \arccos\left(\frac{1}{\sqrt{5}}\right)$ and $B = \arctan\left(\frac{1}{2}\right)$ so that $\cos A = \frac{1}{\sqrt{5}}$ and $\tan B = \frac{1}{2}$

$\therefore \sin A = \frac{2}{\sqrt{5}}$, $\sin B = \frac{1}{\sqrt{5}}$ and $\cos B = \frac{2}{\sqrt{5}}$

So $\sin\phi = \sin(A - B) = \sin A\cos B - \cos A\sin B$

$= \frac{2}{\sqrt{5}} \times \frac{2}{\sqrt{5}} - \frac{1}{\sqrt{5}} \times \frac{1}{\sqrt{5}} = \frac{3}{5}$ as req.

Find the area:

$\frac{1}{2}\int_0^{\phi}(1+\sin\theta)^2 \mathrm{d}\theta + \frac{1}{2}\int_{\phi}^{\frac{\pi}{2}}(2\cos\theta)^2 \mathrm{d}\theta$

Since

$\frac{1}{2}\int (1+\sin\theta)^2 \mathrm{d}\theta = \frac{1}{2}\int (1 + 2\sin\theta + \sin^2\theta)\mathrm{d}\theta$

$= \int \left(\frac{3}{4} + \sin\theta - \frac{1}{4}\cos(2\theta)\right)\mathrm{d}\theta = \frac{3}{4}\theta - \cos\theta - \frac{1}{8}\sin(2\theta)\ (+c)$

And

$\frac{1}{2}\int (2\cos\theta)^2 \mathrm{d}\theta = \frac{1}{2}\int 4\left(\frac{1}{2} + \frac{1}{2}\cos(2\theta)\right)\mathrm{d}\theta = \int 1 + \cos(2\theta)\,\mathrm{d}\theta = \theta + \frac{1}{2}\sin 2\theta\ (+c)$

Then

$\frac{1}{2}\int_0^{\phi}(1+\sin\theta)^2\mathrm{d}\theta+\frac{1}{2}\int_{\phi}^{\frac{\pi}{2}}(2\cos\theta)^2\mathrm{d}\theta$

$=\left[\frac{3}{4}\theta-\cos\theta-\frac{1}{8}\sin(2\theta)\right]_0^{\phi}+\left[\theta+\frac{1}{2}\sin 2\theta\right]_{\phi}^{\frac{\pi}{2}}$

$=\left(\left(\frac{3}{4}\phi-\cos\phi-\frac{1}{8}\sin(2\phi)\right)-(0-1-0)\right)+\left(\left(\frac{\pi}{2}+0\right)-\left(\phi+\frac{1}{2}\sin 2\phi\right)\right)$

$=1+\frac{\pi}{2}-\frac{1}{4}\phi-\cos\phi-\frac{5}{8}\sin(2\phi)$

Since $\sin\phi=\frac{3}{5}$, $\cos\phi=\frac{4}{5}$

Also since $\sin(2\phi)=2\sin\phi\cos\phi$ then $\sin(2\phi)=2\times\frac{3}{5}\times\frac{4}{5}=\frac{24}{25}$

$\therefore\frac{1}{2}\int_0^{\phi}(1+\sin\theta)^2\mathrm{d}\theta+\frac{1}{2}\int_{\phi}^{\frac{\pi}{2}}(2\cos\theta)^2\mathrm{d}\theta$

$=1+\frac{\pi}{2}-\frac{1}{4}\phi-\frac{4}{5}-\frac{5}{8}\times\frac{24}{25}$

$=\frac{\pi}{2}-\frac{2}{5}-\frac{1}{4}\phi$ units^2